
Software Requirements Specification

for

<SG Homie>

Version 1.0 approved

Prepared by:

Steven Octavianus Rahmat

Muhammadh Faheem Faizal

He Hao Yu

Tripathi Eishani

<Group 5, SC2006 FDAF, Nanyang Technological University Singapore>

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Revision History

Name	Date	Reason For Changes	Version
Steven Octavianus Rahmat	April 9, 2025	Completing the documentation	1.0
Muhammadh Faheem Faisal	April 9, 2025	Completing the documentation	1.0
He Hao Yu	April 9, 2025	Completing the documentation	1.0
Eishani	April 9, 2025	Completing the documentation	1.0

1. Introduction

1.1. Purpose

This Software Requirement Specification (SRS) document serves as a guide for developing the SG Homie web application. It provides a clear and detailed outline of the app's requirements, covering everything from key features and interface design to functional and non-functional requirements, as well as any limitations. By documenting these specifications, the SRS helps ensure a smooth development process for all stakeholders involved.

1.2. Document Conventions

This Software Requirements Specification (SRS) document follows standard formatting conventions to maintain clarity and consistency. **Bold** text highlights key points, GUI elements, and user inputs, while *italicized* text indicates newly introduced terms, important notes, or areas that require special attention. Source code, API endpoints, and database queries are displayed in a `monospace` font for easy distinction. Unless explicitly stated otherwise, the priority levels of high-level requirements automatically apply to their detailed counterparts.

Software Requirement Specification Standard: IEEE 830-1998. Priorities of higher level requirements are inherited by detailed level requirements.

Font:	Times New Roman
Heading 1:	Size 19, Bold
Heading 2:	Size 15, Bold
Heading 3:	Size 12, Bold
Heading 4:	Size 12. Bold
Content:	Size 12
Spacing in Content:	1.15-line space

Further conventions on special terms used throughout this document are described in Appendix A: Glossary (Data Dictionary.)

1.3. Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) document is designed to meet the needs of various stakeholders involved in the development and use of **SG Homie**. While reading the entire document provides a comprehensive understanding of the project, we've highlighted the key sections most relevant to each group:

- **Developers** should focus on the system architecture and detailed requirements to understand how to implement the technical aspects.
- **Project Managers** can review the document but should pay close attention to the product scope, objectives, and high-level requirements to ensure everything aligns with project goals.
- **The Marketing Team** should look at the product scope and user interface sections to align their strategies with SG Homie's features and capabilities.
- **End Users** are encouraged to read the overview and user interaction sections to get familiar with how SG Homie works.
- **Testers** should focus on the detailed requirements and exception handling sections to design thorough test plans.
- **Documentation Writers** can use the entire document to create user manuals and help guides.
- **External Developers** who may work with SG Homie's API should refer to the system architecture and technical documentation for integration details.

By structuring the document this way, we aim to make it easier for each stakeholder to find the information they need.

1.4. Product Scope

As Singapore embraces digital transformation, our team introduces SG Homie—a user-friendly web platform designed to simplify the process of finding and analyzing HDB housing options based on individual preferences and real-time market data. SG Homie helps homebuyers, sellers, and investors make informed decisions by providing insights into housing affordability.

The core functionality of SG Homie revolves around its location-based housing search and affordability analysis. By leveraging geolocation services and real estate data, the platform helps users discover available flats, compare resale prices, and evaluate nearby amenities. Users can filter results based on factors like price range, locations, government housing grants, ensuring they find options that best suit their lifestyle and budget.

Beyond just listings, SG Homie provides predictive insights, forecasting price trends, and offering personalized recommendations based on user preferences. The platform also integrates a resale value estimator, helping buyers assess financial feasibility before making a purchase decision.

Additionally, SG Homie allows users to track their housing journey, from browsing listings to saving favorite estates and monitoring price fluctuations. Its intuitive interface presents key information on each HDB estate, including historical pricing trends, accessibility scores, and potential investment value.

With secure authentication and a seamless user experience, SG Homie aligns with Singapore's Smart Nation vision, bringing data-driven, transparent, and accessible housing insights to everyone.

1.5. References

IEE 830-1009 – IEEE Recommended Practice for Software Requirements Specifications. IEEE Standards Association (n.d.). Retrieved 2 April 2025 from <https://standards.ieee.org/ieee/830/1222/>

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Finding a Flat. Housing & Development Board (HDB). (n.d.). Retrieved 2 April, 2025 from <https://homes.hdb.gov.sg/home/finding-a-flat>

2. Overall Description

2.1. Product Perspective

SG Hom-E is a standalone, data-driven real estate platform designed to simplify property search, market analysis, and housing decision-making in Singapore. Aligned with Singapore's Smart Nation vision, the platform integrates geolocation, historical transaction data, and predictive analytics to help users evaluate properties based on price trends, neighborhood amenities (e.g., schools, MRTs), and future valuation scores.

The system offers:

- **Intelligent Property Search:** Filter by location, price range, proximity to amenities, and future price forecasts.
- **Market Analytics:** Interactive charts for historical trends and predictive scoring.
- **User-Centric Features:** Role-based dashboards for buyers, admins, and general users, with secure authentication (Google OAuth).
- **Admin Tools:** User verification, activity monitoring, and data management.

Built with a modular architecture (React.js frontend, Node.js and Express.js backend, Supabase database), SG Hom-E ensures scalability for future integrations (e.g., government housing APIs).

2.2. Product Functions

SG Homie's main functionalities are categorized based on user roles, including administrators and home seekers. These functionalities are further divided into key areas such as User Account Management, Housing Market Insights, Nearby Property Search, and Resale.

2.2.1 User Account Management

1. User Registration
2. User Login
3. User Profile Settings

2.2.2 Property Search

1. Basic/Advanced Filters
2. Map Visualization
3. Like/Interested Property Lists

2.2.3 Market Analytics

1. Historical Price Charts
2. Future Price Predictions
3. Neighborhood Scoring

2.2.4 Admin Controls

1. User Verification
2. Activity Logs
3. Properties Logs
4. Enquiries Logs

2.2.5 Support

1. Contact Form
2. Chatbot for Queries

2.3. User Classes and Characteristics

2.3.1. Buyers

Aspect	Description
Frequency	Frequent
Key Functions Used	Property Search, Market Analytics
Characteristics	Tech-savvy; seeks data-driven decisions; filters by budget, location, amenities.

2.3.2. Unauthenticated

Aspect	Description
Frequency	Occasional
Key Functions Used	Login, Basic Search
Characteristics	New users exploring options; may register after initial searches.

2.3.3. Admin

Aspect	Description
Frequency	Daily
Key Functions Used	User Management, Property Management
Characteristics	Manages system integrity; verifies profiles; resolves disputes.

2.3.4. Seller

Aspect	Description
Frequency	Occasional
Key Functions Used	Add Property
Characteristics	Property owners listing units for sale; Relies on admin for approvals.

2.4. Operating Environment

2.4.1 User Device Permissions

SG Homie requires access to internet connection to operate as all functionalities of the application are based on HTTP requests between the front end, backend, and the database management of the web application. In addition, SG Homie also requires permission from the user to retrieve the location of the device as the location will be used as the benchmark to recommend the nearest hospitals.

2.4.2 Google Permission

SG Hom-E will only request for google profile data if they wish to authenticate via Google OAuth.

2.4.3 Production Environment

SG Hom-E 's frontend interface is hosted with Netlify's cloud hosting service, and for the database management is handled with Google Cloud Console's Firebase. It is compatible with most web-based environments such as Firefox, Safari, Chrome, and Microsoft Edge. It will be platform-independent but optimized for desktop devices.

2.4.3 Development Environment

Development Environment	Description
Front-end Development: React.js (Version 8.2)	React.js is an open-source JavaScript library created by Facebook, designed for building dynamic and responsive user interfaces, particularly for single-page applications. It allows developers to create reusable UI components that can manage and display data in real-time, offering a streamlined approach to building complex and interactive front ends. React's component-based architecture simplifies the development process, as each component manages its own state, enabling more organized and maintainable code. Vite is a modern frontend build tool that provides extremely fast development server start and hot module replacement (HMR). It leverages native ES modules for serving code during development, resulting in near-instantaneous updates without full page reloads. For production builds, Vite uses Rollup under the hood to create highly optimized static assets.
Back-end Development: Node.js (Version 18.17)	Node.js is a powerful, open-source runtime environment built on Chrome's V8 JavaScript engine, allowing developers to run JavaScript on the server side. Its non-blocking, event-driven architecture makes it highly efficient for handling multiple requests simultaneously, making it a popular choice for backend

	development, especially for applications that require real-time data processing.
Database:	Supabase is an open-source backend-as-a-service (BaaS) platform that provides a real-time PostgreSQL database, among other tools, making it especially useful for building mobile and web applications. Its real-time capabilities allow developers to store and sync data across users instantly, which is ideal for applications requiring live updates, such as chat apps, collaborative tools, or social media platforms. With Supabase, data is stored in a structured relational format, making it easy to retrieve and manage while supporting real-time listeners to update the UI instantly when data changes.
Supabase	Beyond its real-time capabilities, Supabase offers extensive support for authentication, analytics, and serverless functions (via Edge Functions), making it an all-in-one solution for developers. Its PostgreSQL foundation enables powerful SQL querying, relational data modeling, and offline support, making it suitable for complex applications. With easy integration into front-end frameworks and compatibility with modern development tools, Supabase is a versatile and scalable choice for developers looking to build dynamic and real-time applications without managing traditional server infrastructure.

2.5. Design and Implementation Constraints

- Budget: Limits advanced API usage (e.g., premium geolocation services).
- Tech Stack: Mandated React/Node.js for consistency.
- Data Compliance: Adheres to PDPA for user privacy.
- Integration Dependency: Relies on Singapore government property APIs for real-time data.

2.6. User Documentation

2.6.1. Introduction

In alignment with Singapore's Smart Nation initiative and the digital transformation of public housing, we present SG Hom-E - an intelligent property platform designed with user-centric principles. By integrating geolocation services with real-time market data, SG Hom-E empowers users to make informed housing decisions based on budget, property type, and location preferences.

The platform's core functionality features an advanced property search system and online application processing. Utilizing sophisticated geocoding technology, SG Hom-E instantly identifies HDB properties near the user while providing intelligent categorization by property type, price range, and neighborhood amenities. Users can seamlessly complete flat viewing appointments and purchase applications through the platform, with all requests being processed by authorized HDB officers for an end-to-end digital experience.

2.6.2. Core Features

- **Intelligent Property Search**
 - SG Hom-E's Smart Search Engine helps users find their ideal HDB flat with precision:
 - Map-Based Search – Browse available properties on an interactive map with real-time filtering.
 - Advanced Filters – Narrow down options by:
 - Price Range
 - Unit Type
 - Location (MRT proximity, school zones, amenities)
 - Future Price Forecast (AI-powered valuation trends)
 - Saved Searches – Store frequently used filters for quick access.
- **Personalized Dashboard**
 - Each user gets a customized homepage with:
 - Favorites List – Save and compare shortlisted properties.
 - Application History – Track past and current submissions.
 - AI Recommendations – Get property suggestions based on search behavior.
 - HDB Officer Admin Portal (Exclusive Access)
- **Authorized HDB personnel can:**
 - Manage Applications – Review, approve, or request additional documents.
 - Adjust Viewing Schedules – Optimize appointment slots based on demand.
 - Generate Reports – Export data on application trends, approval rates, and regional demand.
 - User Management – Monitor registered accounts and resolve issues.

2.6.3. Account Management

2.6.3.1. Login Methods

SG Homie provides two secure login options to meet different user needs.

For users who prefer manual login, they must select the "Login" option on the login page and enter their registered email address and password. The system verifies these credentials in real time, and upon successful authentication, the user is directed to a personalized homepage. If a user forgets their password, the platform sends a secure reset link to their registered email to ensure account security.

For users seeking a more convenient login experience, the platform integrates Google Login. By clicking the "Google Login" button, users can authenticate using their Google account without needing to remember additional credentials. Upon authorization, the platform securely retrieves the necessary identity details, enabling a seamless login experience that automatically directs users to their personal homepage.

3.2 Account Registration

New users can complete their SG Homie account registration in a few simple steps. A prominent "Register" button on the homepage directs users to a step-by-step registration guide.

1. **Basic Information** – Users must provide their legal name, NRIC number, and a valid contact method.
2. **Password Setup** – A strong password must be created, meeting platform security standards (requiring a mix of uppercase letters, lowercase letters, and numbers).
3. **Housing Preferences Survey** – A detailed questionnaire helps the platform understand users' home-buying preferences and needs.
4. **Account Activation** – A verification link is sent to the registered email, ensuring contact information validity before completing the activation process.

2.6.4. Property Search Function

2.6.4.1 Basic Search

SG Homie search system offers three basic search modes to cater to different user preferences:

- **Filter-based Search** – A set of refined filtering options allows users to narrow their search based on price range, property type, and floor area.
- **Pre-set Quick Searches** – The platform provides predefined filters such as "Family-Friendly" or "Convenient Transport" to help users quickly find properties that match specific needs.

2.6.4.2 Advanced Search

For users with more specialized search needs, SG Homie offers an Advanced Search function. Clicking the "Advanced Search" button expands a panel with multiple professional filtering options, including:

- **School Zone Search** – Integrated with official Ministry of Education data, this feature helps users find properties within a 1–2 km radius of top-rated schools.
- **Facility-Based Filtering** – Users can refine their search based on nearby commercial, medical, and recreational facilities, ensuring the property aligns with their lifestyle.

2.6.4.3 Favorites Function

SG Hom-E provides a well-designed Favorites Management System to help users organize preferred properties. Each property listing includes a prominent heart-shaped "Favorite" button. Once clicked, the property is added to the user's Favorites page, displayed in grid or list format. The platform also allows users to create and label custom categories such as "Top Choice," "Backup," or "To Compare," making property management more efficient.

2.6.5. Management Functions (For HDB Officers)

2.6.5.1. User Management

SG Homie provides a comprehensive user management system for HDB officers. After logging into the admin panel, officers can access a complete list of registered users, including basic profile details and recent activity status. Powerful search and filtering tools enable quick identification of specific user accounts. Officers can also adjust user permission levels, such as temporarily restricting application privileges or flagging accounts requiring special attention, ensuring platform security and compliance.

2.6.5.2. Application Review

The platform integrates an intelligent home purchase application review system. When a user submits a new application, the system immediately notifies the responsible HDB officer. Officers can review the full digital application file, with key information automatically extracted and highlighted. The system also performs preliminary checks for completeness and compliance, flagging potential issues requiring manual review. Officers can approve or reject applications online and use the built-in annotation tool to add detailed review notes. All review actions are fully logged for audit tracking.

2.6.5.3. Data Analytics

SG Hom-E's data analytics module provides HDB officers with powerful decision-making support tools. The system automatically generates monthly application reports containing key metrics, such as application volume, approval rates, and regional distribution.

Officers can customize report contents and formats as needed and export reports in Excel or PDF format for presentations or archiving. These data-driven insights help HDB optimize resource allocation and policy-making based on market trends.

2.7. Assumptions and Dependencies

Assumptions:

- Stable internet for real-time data.
- Users grant location permissions.

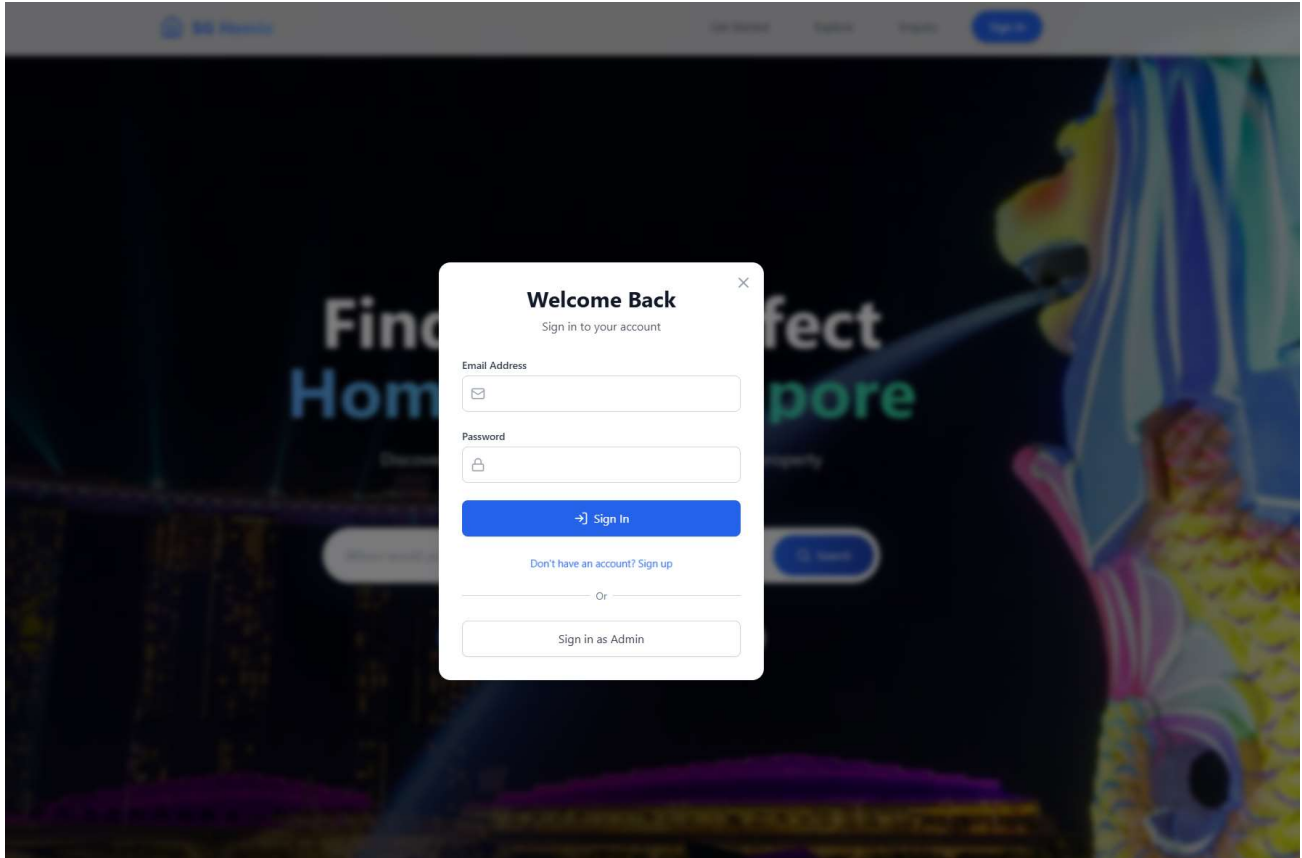
Dependencies:

- Google Maps API for geolocation.
- Firebase uptime for database reliability.

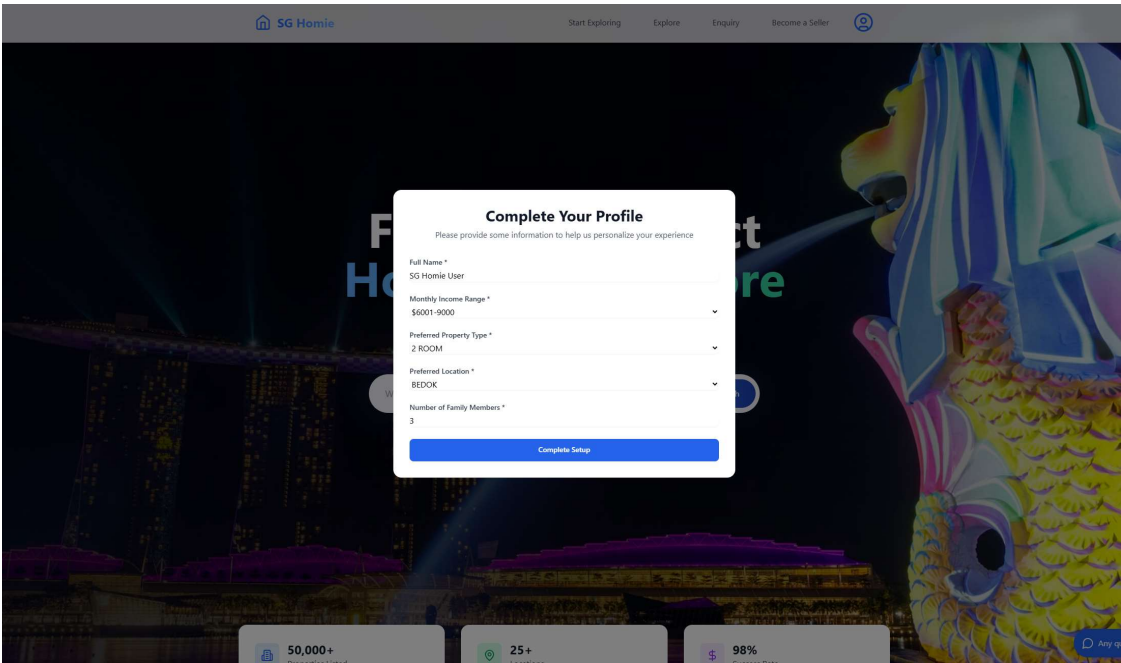
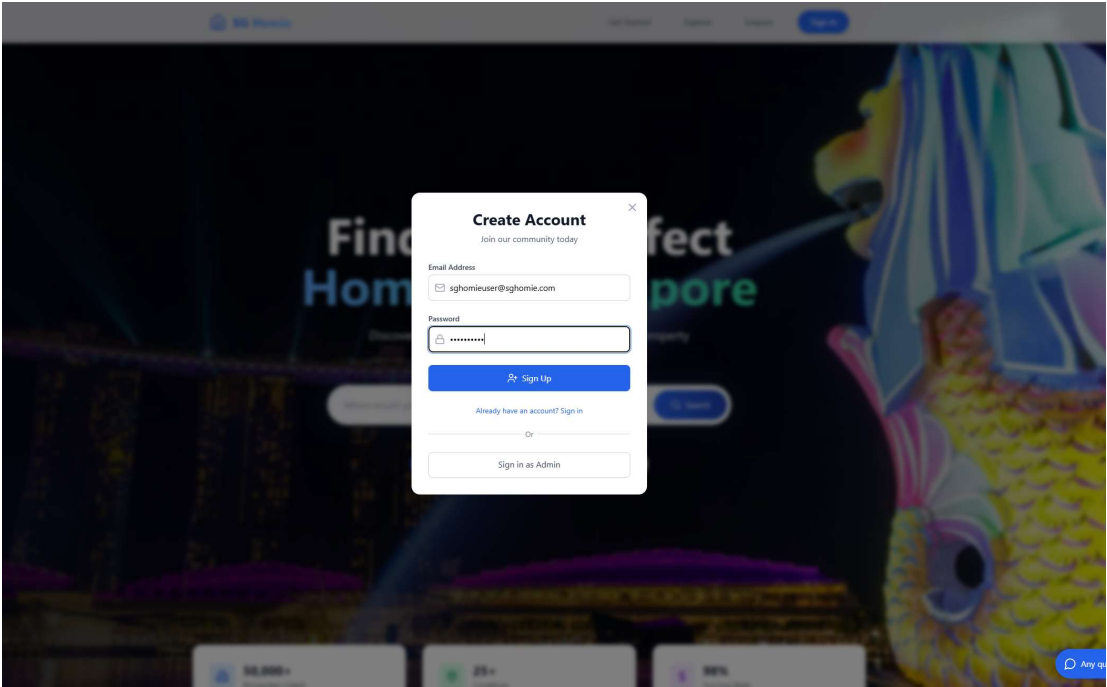
3. External Interface Requirements

3.1. User Interfaces

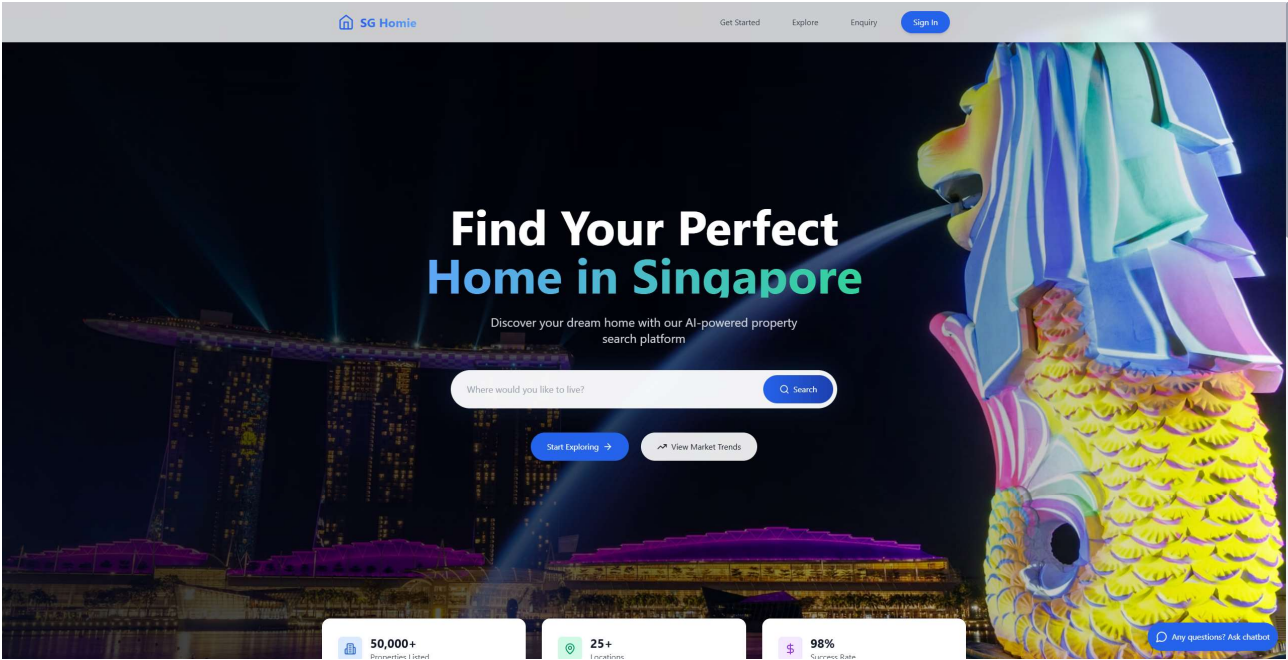
3.1.1. Login Page



3.1.2. SignUp Page



3.1.3. Landing Page



3.1.4. Property Search Page

Start ExploringExploreEnquiryBecome a Seller

Find Your Dream HDB

Search through our curated list of HDB properties across Singapore

☒ Use recommendations from profile

Location

TAMPINES

Room Type

4 ROOM

Minimum Price

\$ Min Price

Maximum Price

\$ Max Price

Search Properties



4 Room HDB

\$550,000

Cozy 4-Room HDB in Tampines

TAMPINES

2 Bathrooms

1,100 sqft

3.1.5. Property Listings Page

Start ExploringExploreEnquiryBecome a Seller



Cozy 4-Room HDB in Tampines

Block 123 Tampines Street 11

\$550,000

\$5500 per sqft

Interested

3 people interested

Rooms

4 Bedrooms

Bathrooms

2 Bathrooms

Floor Area

1100 sqft

Built Year

2010

Description

Beautiful 4-room HDB with excellent amenities nearby



Nearby Amenities

Tampines Mall Shopping

0.3km away

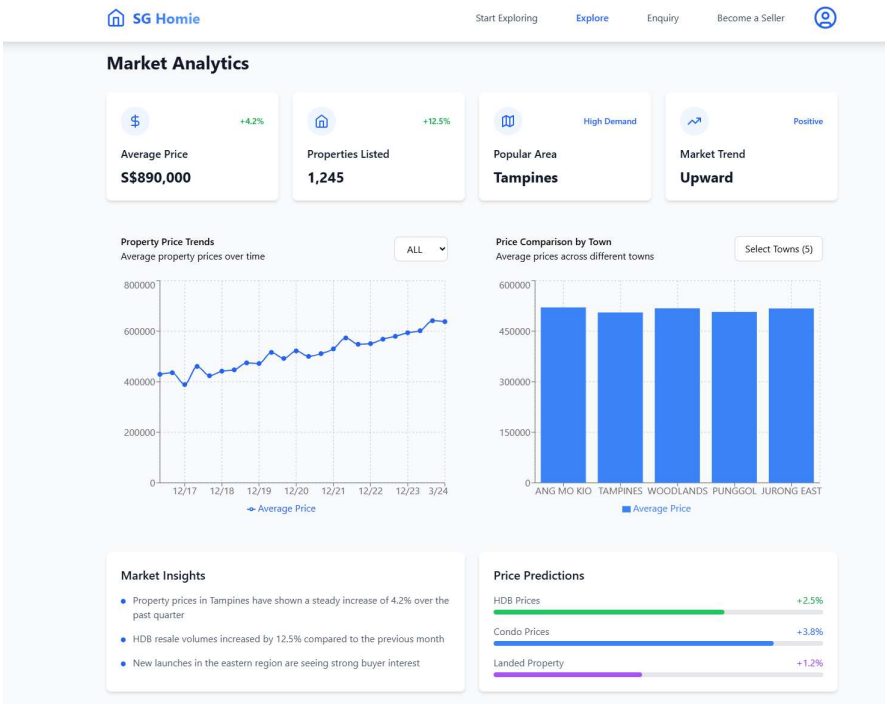
Tampines MRT Transport

0.5km away

Our Tampines Hub Community

0.7km away

3.1.6. Analytics Page



3.1.7. Seller SignUp Page

The 'Become a Seller' registration form is designed to collect essential information from potential sellers. It includes fields for Full Name, Phone Number, Location of Stay, and Email Address. A checkbox for 'I agree to the terms and conditions' is also present. The form is styled with a clean, modern layout and a blue accent color.

Become a Seller

Join our community of property sellers and start listing your properties.

Full Name
SG Homie User

Phone Number
+65 1234 5678
Format: +65 XXXX XXXX

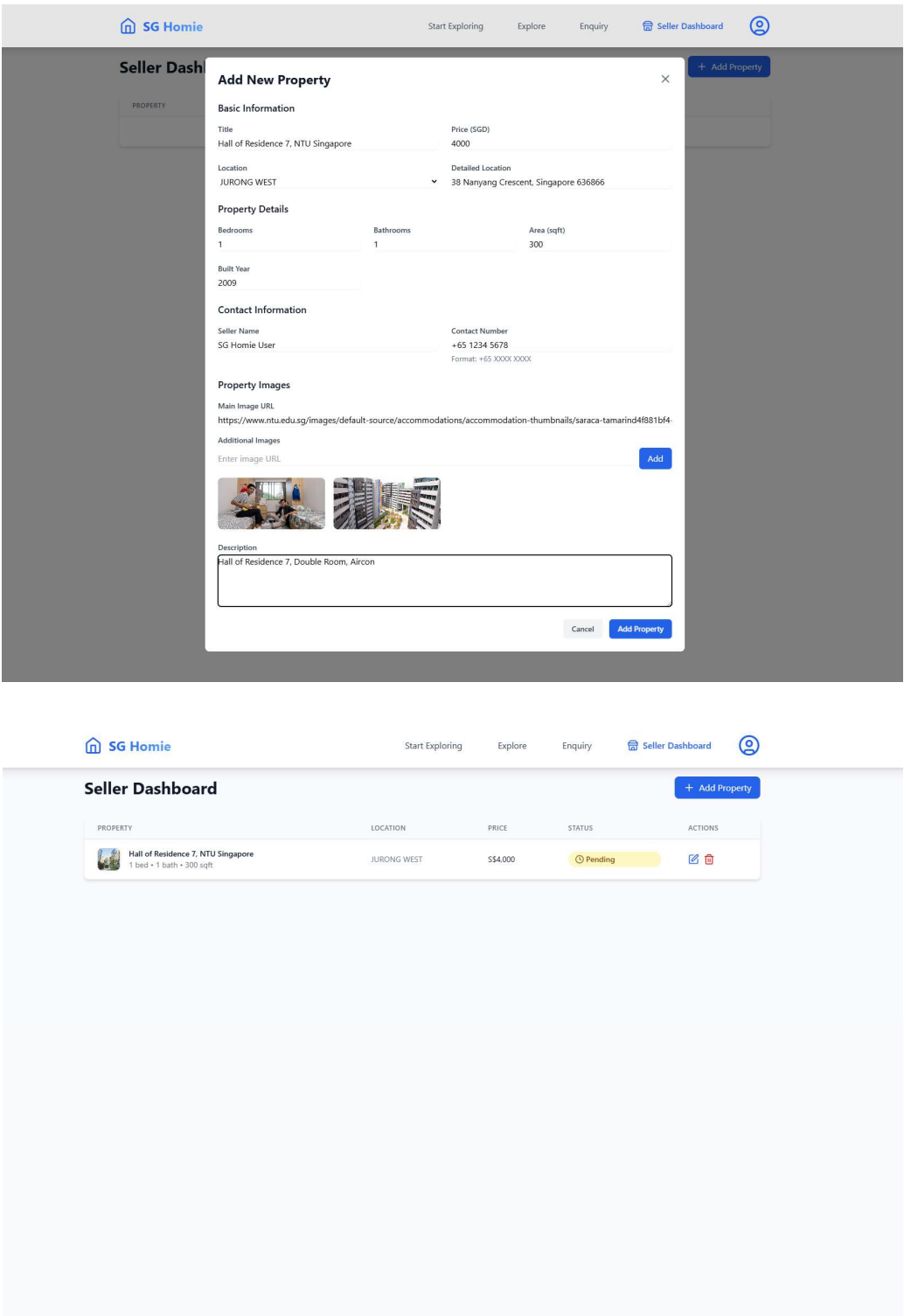
Location of Stay
TAMPINES

Email Address
sghomieuser@sghomie.com

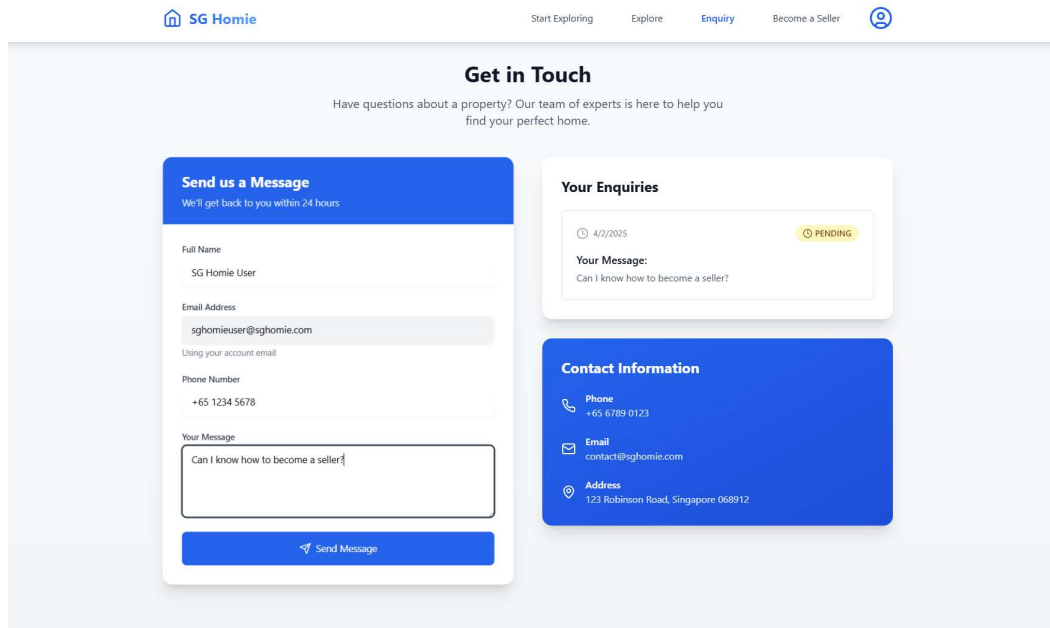
☒ I agree to the terms and conditions
By becoming a seller, you agree to our terms of service and privacy policy.

Register as Seller

3.1.8. Seller Dashboard



3.1.8. Enquiries Page



The screenshot shows the 'Get in Touch' section of the SG Homie website. It features a 'Send us a Message' form on the left and a 'Your Enquiries' section on the right. The form includes fields for Full Name, Email Address, Phone Number, and a message box. The 'Your Enquiries' section shows a pending enquiry with the message 'Can I know how to become a seller?'. Below this is a 'Contact Information' box with phone, email, and address details.

Get in Touch

Have questions about a property? Our team of experts is here to help you find your perfect home.

Send us a Message
We'll get back to you within 24 hours.

Full Name
SG Homie User

Email Address
sghomieuser@sghomie.com

Using your account email

Phone Number
+65 1234 5678

Your Message
Can I know how to become a seller?

Send Message

Your Enquiries

4/2/2025 **PENDING**

Your Message:
Can I know how to become a seller?

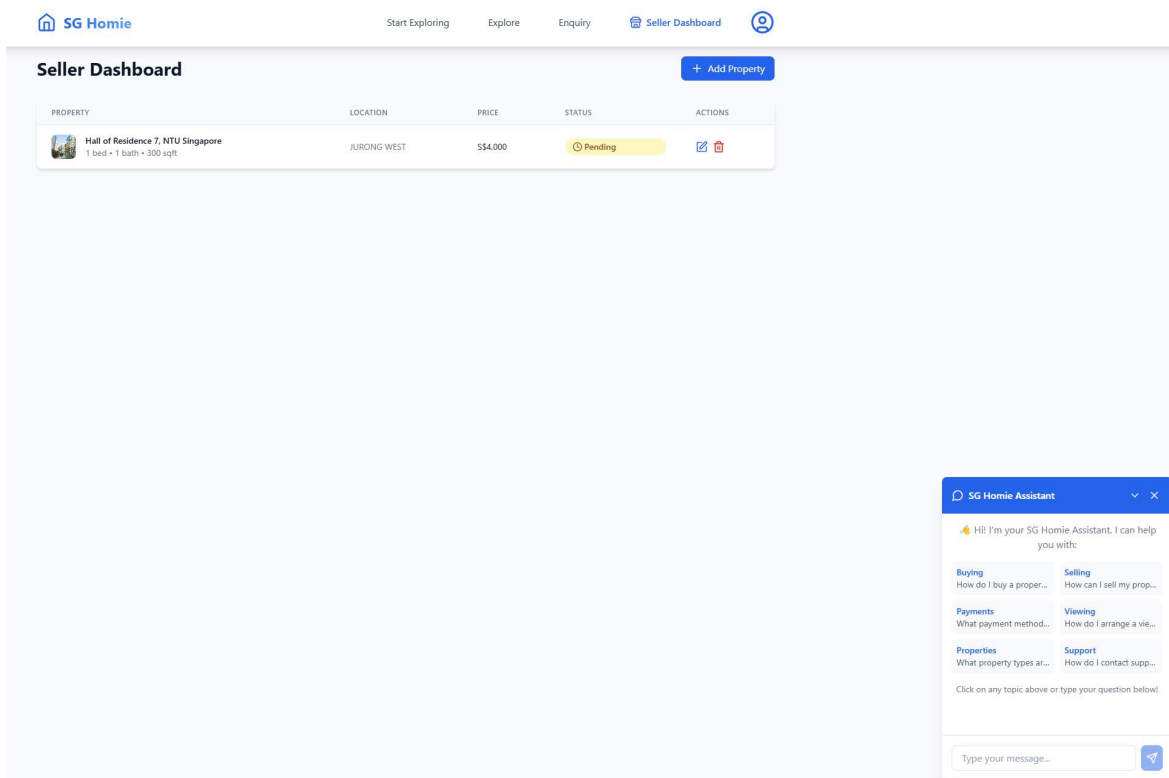
Contact Information

Phone
+65 6789 0123

Email
contact@sghomie.com

Address
123 Robinson Road, Singapore 068912

3.1.9. Chatbot



The screenshot shows the 'Seller Dashboard' of the SG Homie website. It includes a table with property listings and a chatbot window at the bottom right. The table has columns for Property, Location, Price, Status, and Actions. The chatbot window, titled 'SG Homie Assistant', displays a greeting and a list of topics: Buying, Selling, Payments, Viewing, Properties, and Support. A text input field for the chatbot is also visible.

Seller Dashboard

PROPERTY

PROPERTY	LOCATION	PRICE	STATUS	ACTIONS
 Hall of Residence 7, NTU Singapore 1 bed • 1 bath • 300 sqft	JURONG WEST	\$54,000	Pending	 

SG Homie Assistant

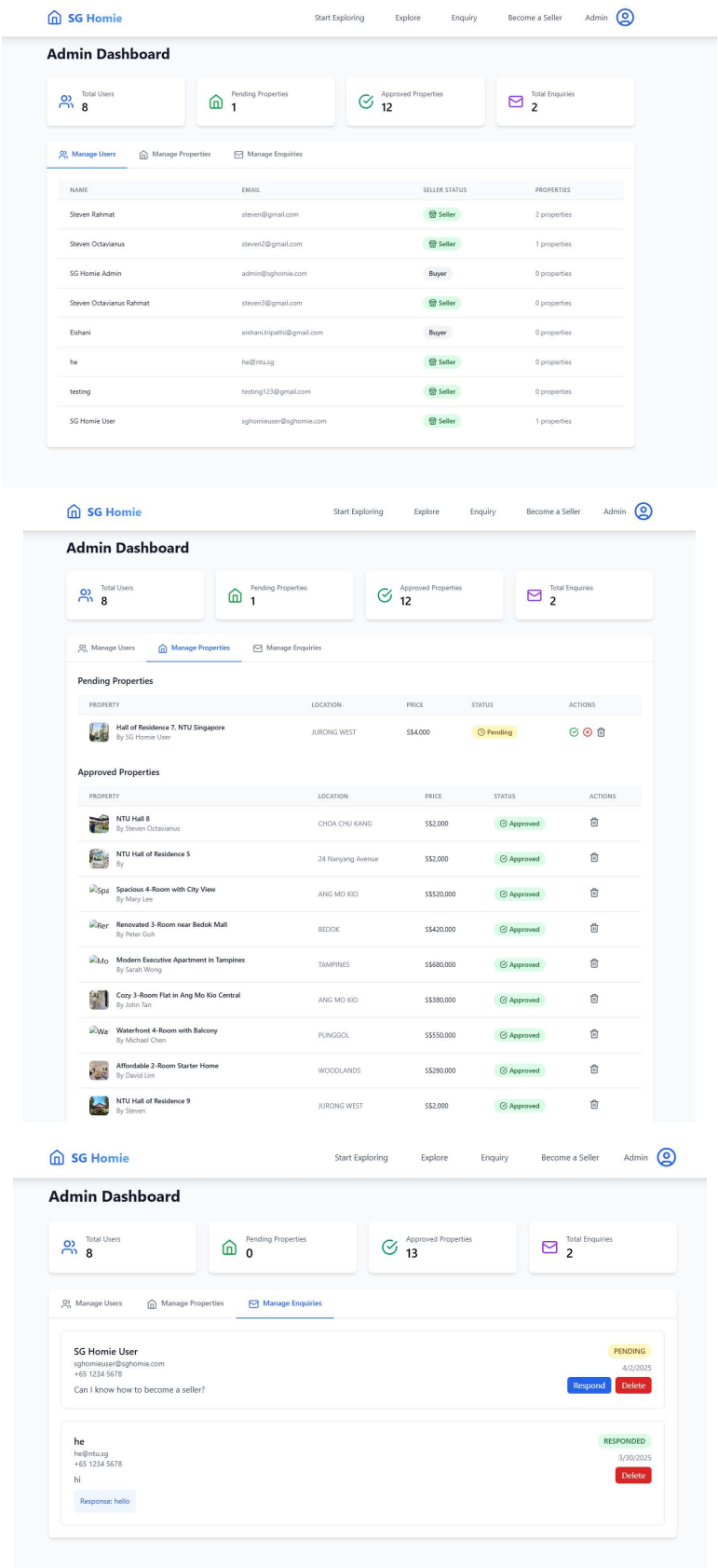
Hi! I'm your SG Homie Assistant. I can help you with:

- Buying**
How do I buy a proper...
- Selling**
How can I sell my prop...
- Payments**
What payment method...
- Viewing**
How do I arrange a vie...
- Properties**
What property types ar...
- Support**
How do I contact supp...

Click on any topic above or type your question below!

Type your message...

3.1.10. Admin Dashboard



3.2. Hardware Interfaces

This section describes the hardware interfaces required for SG Homie web application to perform *effectively and reliably*.

Requirements	Description
Operating System	- For Windows: Windows 8 and later versions. - For macOS: macOS 10.10 and later versions. - For Linux: Ubuntu, Debian, Fedora
Network Connection	Wireless Network Interface Card (WNIC) or a modern chip with cellular modem.
Interaction	A functional touchpad or mouse to navigate through the website

3.3. Software Interfaces

OS : Windows 11

Tools : ReactJS, NodeJS, ExpressJS, Supabase

Third-party libraries:

- React & React DOM:
 - o The core of the frontend is built using React, with components such as Navbar, Hero, Chatbot, etc.
- React Router DOM:
 - o Used for client-side routing across pages (e.g., Home, Analytics, PropertyDetails, etc. in the src/pages/ directory).
- Vite:
 - o The project uses Vite as its build tool and development server. Configuration is provided in vite.config.ts.
- Tailwind CSS:
 - o For styling, the project uses Tailwind CSS. The configuration is in tailwind.config.js, and PostCSS is configured via postcss.config.js.
- Recharts:
 - o Used for creating charts and visual data representations, likely on pages such as Analytics.tsx.
- React Markdown
 - o This library is used to render markdown content within React components.
- Lucide React:
 - o An icon library for React components.
- Leaflet and React Leaflet:
 - o An icon library for React components.

- Axios:
 - An icon library for React components.

APIs:

- OpenStreetMap API
 - The OpenStreetMap API provides developers with access to OpenStreetMap's rich, crowd-sourced geographic data, allowing them to retrieve, modify, and integrate detailed mapping and location information into their applications.
- HDB Price Listings API:
 - The HDB Price Listings API provides HDB selling price from 2017 – onwards, which is used to fetch the data for the Machine Learning models.

3.4. Communications Interfaces

To ensure smooth functionality, especially for account registration and property finding, the website will communicate with users via the email they registered with. For property finding, users will need to complete their personal profile and preferences.

Since the platform relies on real-time data, a stable internet connection is required to fetch information from APIs and the user database. This includes actions like GET, PATCH, DELETE through HTTP requests. Without an internet connection, some features may not function properly, as the system won't be able to fetch the required data.

4. System Features

This section covers the core system features of SG Homie

4.1. User Authentication and Profile Management

4.1.1 Description and Priority

Overall Priority	Description
High	Logging in is the gateway to all restricted functionalities on the platform. Without successful login, the user cannot perform actions like viewing listings and sending enquiries

4.1.2 Stimulus/Response Sequences

Use Case ID:	#1 – 1		
Use Case Name:	User Login		
Created By:	Faheem	Last Updated By:	Faheem
Date Created:	08/04/2025	Date Last Updated:	08/04/2025

Actor:	User
Description:	- This use case describes how a user (seller, buyer, admin) accesses the system by providing valid login credentials (e.g., email/username and password). Once authenticated, the user can access all authorized functionalities, such as viewing or editing properties, submitting enquiries, etc.
Preconditions:	- The User has already signed up with their credentials - The User is connected to the internet
Postconditions:	- The User is successfully authenticated and is directed to the Dashboard - The User remains unauthenticated if credentials are invalid, and an error message, “Invalid login credentials” is displayed
Priority:	High
Frequency of Use:	High frequency, users must log in to access any features
Flow of Events:	1. Users navigates to the login page 2. User inputs valid credentials and submits the login form

	3. Login page forwards the credentials to the Authentication controller 4. Authentication controller queries Account database and verifies user's credentials 5. Account database returns user record details if credentials match [true], else, returns [false] 6. Authentication controller sends a success status to Login page if verification is valid 7. Login page displays a success message sets the user's session and redirects them to the SGHomie Dashboard
Alternative Flows:	1. Users navigates to the login page 2. User inputs valid credentials and submits the login form 3. Login page forwards the credentials to the Authentication controller 4. Authentication controller queries Account database and verifies user's credentials 5. Account database indicates that the credentials are invalid (User does not exist or wrong email/password) and returns [false] 6. 7. Authentication controller returns a login failure status to the Login page 8. Login page displays error message "Invalid credentials" and prompts the user to try again.
Exceptions:	
Includes:	
Special Requirements:	Consider secure password handling
Assumptions:	- The user's device can display the Login Page properly
Notes and Issues:	

Use Case ID:	#1 – 2		
Use Case Name:	User Create Account		
Created By:	Faheem	Last Updated By:	Faheem
Date Created:	08/04/2025	Date Last Updated:	08/04/2025

Actor:	User
Description:	- This use case describes how a user (seller, buyer, admin) creates an account in SGHomie by providing

	personal details (valid email address and password). Upon successful sign-up, the user can log in and access platform features.
Preconditions:	- The user does not already have an SGHomie account. - The user has access to the 'Sign Up' page or function. - The user has a valid email address
Postconditions:	- The User successfully creates an account and is directed to 'Complete Profile' section - The User fails to create an account if credentials are invalid such as duplicate email or invalid input and an error message is displayed
Priority:	High - Account creation is essential to access personalized or restricted features
Frequency of Use:	Medium – A user typically signs up one time, but new users may sign up continually
Flow of Events:	1. User navigates to the 'SignUp' form on SGHomie (top-right of the page) 2. User fills in registration details (Valid email address and password) 3. System (Sign-up form) validates basic input client-side (e.g., required fields such as '@' symbol for email address). 4. System sends the registration data to the 'SignUp' service (Account Database). 5. Sign Up service checks for duplication (e.g., existing email). If no duplicates, it stores the new user data in the database. 6. Database returns a success status. 7. Sign Up service notifies the Sign Up form of the successful registration.
Alternative Flows:	
Exceptions:	1. Network error preventing the sign-up service from storing data - o System shows "Unable to create account. Please try again later." 2. Duplicate email - o System displays "User already registered"
Includes:	None
Special Requirements:	System should securely store and handle password

Assumptions:	- The user's device can display the Login Page properly - The user has a stable internet connection
Notes and Issues:	- Two-factor authentication (2FA) might be considered in the future. - Real-time validation or client-side checks for phone/email formatting could be added

Use Case ID:	#1 – 3		
Use Case Name:	Complete Profile		
Created By:	Faheem	Last Updated By:	Faheem
Date Created:	08/04/2025	Date Last Updated:	08/04/2025

Actor:	User
Description:	- This use case describes how a logged-in user (seller, buyer, admin) updates or completes their SGHomie profile details, such as monthly income range, job location, number of family members, preferred property type and location preferences to refine the platform's recommendations
Preconditions:	- The user is logged in to SGHomie (i.e., valid account and session). - The user has partial or empty profile data that they want to complete/update. - The user is on the Profile Completion page or function
Postconditions:	- The user's profile is successfully saved in the system, ensuring that their data (income range, preferred location, etc.) is up to date - The user fails to create profile if the mandatory profile fields are missing or the system encounters a save error, the user is prompted to correct or retry
Priority:	High - Properly completed profile data is crucial for personalized recommendations of properties
Frequency of Use:	Medium – A user might update their profile occasionally or whenever personal circumstances change (income, family size, etc.)
Flow of Events:	1. User is prompted to the “Complete your Profile” page after successful account creation

	2. User fills in and selects the appropriate dropdown options in the relevant profile fields (Full name, monthly income range, number of family members, preferred location, preferred property type). 3. System (UI) validates basic input client-side 4. System sends the updated data to the Profile Management Service (backend). 5. Profile Management Service checks for completeness. If required data is missing, it responds with an error message. 6. Upon successful validation, the Profile Management Service stores the new data in the User Profile Database. 7. The database returns a success acknowledgment. 8. The Profile Management Service confirms success to the user interface. 9. The User Interface displays a “Profile successfully updated” message.
Alternative Flows:	None
Exceptions:	1. Network error preventing database access ○ System shows error prompt “Unable to update profile, please try again later 2. Required fields not provided ○ System shows error prompt “Please fill in the required fields
Includes:	Verify User – The user must be authenticated before accessing or updating profile details
Special Requirements:	- Input validation for numeric fields - Potential phone/email re-validation if the user changes them
Assumptions:	- The user’s device can display the Profile Page properly - The user has a stable internet connection
Notes and Issues:	- The system can enforce certain constraints on the numeric range for income or family members (e.g., $0 < \text{family_members} \leq 15$)

4.1.3 Functional Requirements

1. User Authentication & Profile Management

1.1. SG Hom-E shall allow users to create an account.

1.1.1. Users shall provide their name, email, password.

- 1.1.2. Users shall make a password with a minimum of 6 characters.
- 1.2. SG Hom-E shall allow users to log in using email and password.
 - 1.2.1. SG Hom-E shall allow users to enter their email and password.
 - 1.2.2. SG Hom-E shall tell that existing users had existed, or wrong credentials is entered.
- 1.3. SG Hom-E shall allow users to update their profile information.
 - 1.3.1. Users can update the profile with:
 - 1.3.1.1.Number of rooms needed (3 rooms, 4 rooms, ...)
 - 1.3.1.2.Income range
 - 1.3.1.3.Job location
 - 1.3.1.4.Preferred Location for Housing
 - 1.3.1.5.Number of Family Member
 - 1.3.2. Users can set notification preferences for market updates and property alerts.

4.2. Property Search & Recommendation System

4.2.1. Description and Priority

Overall Priority	Description
High	Enable users (buyers or renters) to search for properties based on personalized filters such as location, price range, property type, floor area, proximity to MRT, schools, or other lifestyle preferences. The system also uses AI-driven recommendation algorithms to suggest suitable properties based on user profiles. Results are presented with sorting, bookmarking, and comparison features to enhance the decision-making process.

4.2.2. Stimulus/Response Sequences

Use Case ID:	#2 – 1		
Use Case Name:	QueryProperties		
Created By:	Steven	Last Updated By:	Steven
Date Created:	9 April 2025	Date Last Updated:	9 April 2025

Actor:	User
Description:	This use case allows a registered user to search for properties based on various criteria such as location, price range, property type, floor area, and proximity to amenities. The system retrieves and displays a list of properties matching the user's criteria, enabling them to view detailed information, save favorites, and initiate contact with sellers.
Preconditions:	The user must be registered and logged into the system.
Postconditions:	- The user has viewed a list of properties matching their search criteria. - The user may have initiated contact with property sellers.
Priority:	High
Frequency of Use:	High. Users are expected to use the property search functionality frequently to find suitable properties.
Flow of Events:	1. The user should login to their registered account 2. The user should already complete their profile page 3. The user accesses the property search page

	4. The user submits the request 5. The system retrieves the matching properties listing from the database 6. The system displays the list of matching properties
Alternative Flows:	If the system doesn't find one, then it will show "No Properties found"
Exceptions:	If user is not logged in, they will be redirected to the main page.
Includes:	FilterProperties
Special Requirements:	None
Assumptions:	• Users have a basic understanding of how to use search interfaces. • Property data in the system is accurate and up to date.
Notes and Issues:	None

Use Case ID:	#2 – 2		
Use Case Name:	FilterProperties		
Created By:	Steven	Last Updated By:	Steven
Date Created:	9 April 2025	Date Last Updated:	9 April 2025

Actor:	User
Description:	This use case allows users to refine property search results by applying additional filters such as property type, number of rooms, price range. It enhances discoverability and ensures the search results align closely with the user's needs and lifestyle preferences.
Preconditions:	• The user is logged in. • The user has initiated a property search or is viewing property listings.
Postconditions:	A refined list of properties is displayed based on the selected filters.
Priority:	High
Frequency of Use:	Frequent – especially right after a general search, when users want to narrow down a long list of results.
Flow of Events:	1. After the property search, user might want to change the filter of the properties, like the other place they

	actually want the price range, and the type-unit of the flat they want 2. The user adjusts the filter 3. The user clicks on “Search Properties” again 4. A refined list of property listings is shown, sorted according to relevance or user-selected order (e.g., price low-to-high).
Alternative Flows:	If the system doesn’t have any results after filtering, then it will display “No Properties found”
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.2.3. Functional Requirements

2. Property Search & Recommendation System

2.1. SG Hom-E shall allow users to search for resale and new properties.

2.1.1. Users can filter properties by location, price range and property type.

2.1.2. The system shall display search results with price trends, accessibility scores, and locations.

2.2. SG Hom-E shall provide property recommendations.

2.2.1. Recommendations shall be based on user profile data or based on the user’s filter preferences.

2.2.2. Users shall receive explore details about the properties based on their preferences.

4.3. Market Analytics and User Insights

4.3.1. Description and Priority

Overall Priority	Description
High	Provides users with data-driven insights into the property market using interactive visualizations, charts, and AI-powered predictions. Users can explore trends in property prices, transaction volumes, Personalized insights are also generated based on user profiles, helping them make more informed decisions.

4.3.2. Stimulus/Response Sequences

Use Case ID:	#3 – 1		
Use Case Name:	View Analytics		
Created By:	He Hao Yu	Last Updated By:	He Hao Yu
Date Created:	9 April 2025	Date Last Updated:	9 April 2025

Actor:	User, System
Description:	SG Homie displays comprehensive market analytics comparing property trends across selected towns.
Preconditions:	- User is logged in - Historical market data is available - Analytics module is properly integrated
Postconditions:	- System displays comparative analytics visualization - User can export selected data
Priority:	High
Frequency of Use:	Medium
Flow of events:	1. User navigates to Analytics page 2. System loads default comparison (last 5 years) 3. User selects up to 5 towns for comparison 4. User adjusts date range (min 2010, max current year) 5. System retrieves and aggregates historical data 6. System generates interactive line/bar charts 7. System displays comparative analysis
Alternative Flows:	None

Exceptions:	None
Includes:	View Market Insights, View Past Future Data
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	#3 – 2		
Use Case Name:	View Market Insights		
Created By:	Steven	Last Updated By:	Steven
Date Created:	9 April 2025	Date Last Updated:	9 April 2025

Actor:	User
Description:	This use case enables users to view up-to-date market statistics and insights on the current state of the Singapore property market. It provides key metrics like average prices, volume changes, market demand, popular locations, trend directions, and price predictions across housing types, HDB. These insights are presented in a visually appealing, dashboard-style format to help users quickly understand where the market is heading.
Preconditions:	The user is logged in
Postconditions:	• The user can view the market analytics. • The user has a clear, high-level understanding of the current property market. • The user may choose to explore deeper analytics or start filtering/searching properties.
Priority:	Medium
Frequency of Use:	High – users frequently return to this dashboard to monitor the market or check if it's the right time to buy, sell, or invest.
Flow of Events:	1. The user clicks on the Analytics button 2. The user can now view the market analytics
Alternative Flows:	None
Exceptions:	None
Includes:	None

Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	#3 – 3		
Use Case Name:	View Past and Prediction		
Created By:	Steven	Last Updated By:	Steven
Date Created:	9 April 2025	Date Last Updated:	9 April 2025

Actor:	User
Description:	This use case allows users to view historical predictions made by the platform's AI/ML models alongside actual outcomes. Users can compare predicted versus actual property price trends, transaction volumes, and market movements for specific estates, time periods, or policy events. This retrospective view increases trust in the platform's forecasting capability and helps users evaluate model reliability before making decisions.
Preconditions:	The user is logged in
Postconditions:	- The user has visualized the accuracy of previous predictions. - The user may have selected regions, dates, or metrics to compare.
Priority:	High
Frequency of Use:	Moderate – used by data-conscious users or during evaluation of system trustworthiness.
Flow of Events:	- The user clicks on the Analytics button - The user chooses which year they want to see the past prices - The user chooses up to 5 towns that they want to compare the prices
Alternative Flows:	None
Exceptions:	None
Includes:	None

Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.3.3. Functional Requirements

3. Market Analytics & Trend Forecasting

3.1. SG Homie shall provide market analytics & price predictions

3.1.1. SG Homie shall provide an Analytics Page that displays historical housing market data and predictive price trends.

3.1.2. Users shall be able to view and compare past property prices across different towns.

3.1.3. The system shall provide future price forecasts based on market trends and predictive models.

3.2. SG Homie will give property price comparison

3.2.1. Users shall be able to compare property prices across different towns.

3.2.2. Users can select a custom date range (e.g., 2010–2024) for comparison.

3.2.3. A maximum of five towns can be selected for direct comparison on the analytics page.

3.2.4. The system shall display a graphical comparison of price trends across selected towns.

3.3. SG Hom-E shall provide market insights & notifications

3.3.1. SG Homie shall provide users with market insights and analysis related to housing trends.

4.4. Neighborhood Insights & Location Based Features

4.4.1. Description and Priority

The user can find insights about neighborhoods and nearby facilities.

Overall Priority	Description
High	Provides users with comprehensive insights about neighborhoods and lifestyle considerations by facilitating searches for nearby places, public facilities, schools, and housing scores, assisting users in evaluating property locations and their surrounding amenities effectively.

4.4.2. Stimulus/Response Sequences

Use Case ID:	#4 - 1		
Use Case Name:	DisplayPropertyOnMap		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	User, System
Description:	SG Homie displays property locations on an interactive map using OneMap API to assist users in visualizing properties and nearby amenities.
Preconditions:	1. Property location data is available in the system. 2. OneMap API is integrated with SG Homie.
Postconditions:	The system displays the property on the map, along with nearby amenities and their distances.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. User selects a property listing. 2. The system retrieves location data for the selected property. 3. The system uses OneMap API to display the property location on an interactive map. 4. Nearby amenities such as MRT stations, schools, malls, and hospitals are highlighted. 5. The system calculates and displays distances to key amenities.

Alternative Flows:	If location data is unavailable, the system notifies the user and provides a link to contact admin for assistance.
Exceptions:	None
Includes:	CalculateDistancesUsingAPI
Special Requirements:	None
Assumptions:	OneMap API is functioning optimally.
Notes and Issues:	None

Use Case ID:	#4-2		
Use Case Name:	DisplayPropertyDetails		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	User
Description:	This use case involves a user searching for various types of places within a specified location or area. The system will return a list of places that match the user's criteria, helping them to identify and choose locations for their specific needs.
Preconditions:	1. Seller has uploaded complete property details and images. 2. System has verified and approved property data.
Postconditions:	Users can view all relevant details about a property, including seller contact information for inquiries
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User selects a property listing. 2. The system displays property images, descriptions, and location data. 3. Seller details, including name and contact information, are provided for inquiries. 4. The user can click on the seller's contact info to initiate communication.

Alternative Flows:	If seller contact details are unavailable, the system notifies the user and provides an option to report the issue to admin.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.4.3. Functional Requirements

4. Neighborhood Insights & Location-Based Features

4.1. SG Homie shall provide property location and mapping

4.1.1. SG Homie shall display the property's location on an interactive map using the OneMap API.

4.1.2. Users shall be able to view nearby amenities such as MRT stations, schools, malls, and hospitals.

4.1.3. The system shall calculate and display distances to key amenities from the property location.

4.2. SG Homie will give information about the seller.

4.2.1. SG Homie shall display property images to give users a visual representation of listings.

4.2.2. Each listing shall include seller details such as contact information for inquiries.

4.3. SG Homie will be integrated with OneMapAPI

4.3.1. SG Homie shall integrate the OneMap API to provide geospatial mapping of property locations.

4.3.2. The system should use OneMap API for distance calculations between properties and essential amenities.

4.3.3. Users shall be able to explore the surrounding area of a property directly on the map.

4.5. Property Listing & Seller Management

4.5.1. Description and Priority

Overall Priority	Description
High	Enables sellers or property agents to create, update, and manage property listings on the platform. Each listing includes details like location, price, property type, floor area, images, nearby amenities, and availability. Listings are searchable and filterable by buyers based on criteria like location, price range, or property features.

4.5.2. Stimulus/Response Sequences

Use Case ID:	#5 – 1		
Use Case Name:	Register Seller		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	Seller
Description:	Seller registers on SG Homie by providing their personal information to create an account
Preconditions:	Seller must have account and access to SG Homie's registration page.
Postconditions:	Sellers successfully register and receive confirmation of account creation.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Sellers navigate to SG Homie's registration page. 2. Sellers enter their full name, mobile number, and email address into the required fields. 3. Sellers confirm their details and submit the form. 4. The system validates the data and creates the seller's account 5. User is directed to seller dashboard
Alternative Flows:	If all the details are not filled, the system displays an error message requesting the user to complete all fields.
Exceptions:	None

Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	#5 – 2		
Use Case Name:	Add Property Listing		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	Seller
Description:	Sellers add new property listings by providing detailed information and uploading images.
Preconditions:	Seller must be logged in and registered as a seller.
Postconditions:	Property listing is successfully submitted and awaits admin approval.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Seller logs into their account. 2. Seller navigates to the "Add Property Listing" section. 3. Seller enters details like property title, asking price, location, etc. 4. Seller uploads property images and adds a description. 5. Seller submits the listing.
Alternative Flows:	If mandatory fields are not filled, the system displays an error prompting the seller to complete the form.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	Seller has high-quality images of the property.
Notes and Issues:	None

Use Case ID:	#5 – 3		
Use Case Name:	VerifyPropertyListing		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	Admin
Description:	The admin reviews property listings submitted by sellers for approval or rejection.
Preconditions:	Property listing has been submitted by a seller. Admin is logged into the system and authorized to verify listings.
Postconditions:	Approved properties are visible to the public, and sellers are notified.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Admin logs into the system using valid credentials. 2. Admin navigates to the "Verify Property Listings" section. 3. Admin reviews a listing and either approves or rejects it. 4. The system sends an email to the seller notifying them of the decision.
Alternative Flows:	If the listing contains incomplete information, admin marks it as "Incomplete," and the system notifies the seller to update details.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	Admin reviews listings.
Notes and Issues:	None

4.5.3. Functional Requirements

5. Property Listing & Seller Management

5.1. SG Homie will allow users to register as a seller

- 5.1.1. SG Homie shall allow users to **register as sellers** by completing their profile information.
 - 5.1.2. Sellers must provide:
 - 5.1.2.1.Full name
 - 5.1.2.2.Mobile number
 - 5.1.2.3.Email address
 - 5.1.3. After successful registration, sellers shall be able to list their properties for sale.
- 5.2. Seller should be able to add property listing.
- 5.2.1. Sellers shall be able to add new property listings by providing:
 - 5.2.1.1.Property title/name
 - 5.2.1.2.Asking price
 - 5.2.1.3.Location (town, estate)
 - 5.2.1.4.Number of bedrooms and bathrooms
 - 5.2.1.5.Floor area (in sqm)
 - 5.2.1.6.Year built
 - 5.2.1.7.Seller's name and mobile number
 - 5.2.1.8.Link to property images
 - 5.2.1.9.Description of the property
 - 5.2.2. Sellers shall be able to edit or remove their listings before admin approval.
- 5.3. Properties should be verified before being available to the public.
- 5.3.1. All submitted properties shall require approval from SG Homie Admin before being published.
 - 5.3.2. Once approved, the property shall be visible to the public in the listings section.
 - 5.3.3. The system shall notify the seller via email when their listing is approved or rejected.

4.6. User Engagement and Support

4.6.1. Description and Priority

The user can ask queries to the chatbot which responds and submit enquiries which the admin answers.

Overall Priority	Description
Medium	Provides chatbot assistance to help users with property searches, financing questions, and market trends, while also providing an Enquiry Page for submitting inquiries when chatbot responses are insufficient, allowing users to input their details, view past inquiries, and receive replies from admins directly.

4.6.2. Stimulus/Response Sequences

Use Case ID:	#6-1		
Use Case Name:	User Enquiries		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	User
Description:	Users submit inquiries through the Enquiry Page when the chatbot is unable to address their questions.
Preconditions:	User must be logged into their account.
Postconditions:	The enquiry is successfully submitted, and the user can view it alongside their past enquiries.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User logs into their SG Homie account. 2. User navigates to the Enquiry Page. 3. User fills in the enquiry form, providing full name, email address, phone number, and a description of the enquiry. 4. User reviews the information entered and submits the form.

	- The system validates the input and saves the enquiry in the database. - The enquiry appears in the user's past enquiries list, displayed beside the form
Alternative Flows:	If required fields are left blank or invalid data is entered, the system displays an error prompting correction.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Consider implementing notifications to inform users when admin replies to their enquiries.

Use Case ID:	#6-2		
Use Case Name:	ReplyUserEnquiries		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	Admin
Description:	Replying to the Chat from the users if they have additional enquiries that can't be answered by the chatbot
Preconditions:	Admin must have valid login credentials and proper authorization.
Postconditions:	Admin successfully initiates the selected activity.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	- The admin is successfully logged in into the dashboard - The admin selects Reply User Enquiries - The admin clicks on the user they want to reply

	4. The admin replies their messages with relevant information to the user.
Alternative Flows:	If the admin lacks the required permissions for a specific task, the system displays an error message.
Exceptions:	None
Includes:	Verify User
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.6.3. Functional Requirements

6. User Engagement & Support

6.1. SG Homie will provide chatbot assistance for users.

6.1.1. SG Homie provides a chatbot to assist users with property search, financing queries, and market trends.

6.1.2. Users shall be able to request human support if chatbot responses are insufficient.

6.2. SG Homie provides an Enquiry system for users to ask if chatbot can't answer their question.

6.2.1. SG Homie shall provide an **Enquiry Page** where users can submit inquiries.

6.2.2. Users shall be required to input:

6.2.2.1.Full name

6.2.2.2.Email address

6.2.2.3.Phone number

6.2.2.4.Description of the enquiry

6.2.3. Users shall be able to **view their past enquiries** beside the form.

6.2.4. The system shall allow users to **view replies from the admin** once a response has been provided.

4.7. Admin Management System

4.7.1. Description and Priority

Overall Priority	Description
High	Provides an Admin Management System that enables admins to oversee user accounts, manage property listings, and handle user enquiries effectively, ensuring proper authorization, reviewing submissions, and maintaining system integrity

4.7.2. Stimulus/Response Sequences

Use Case ID:	#7 – 1		
Use Case Name:	SelectAdminActivity		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	Admin
Description:	The admin selects and initiates administrative functions, such as managing user data, updating system configurations, or reviewing reports.
Preconditions:	Admin must have valid login credentials and proper authorization.
Postconditions:	Admin successfully initiates the selected activity.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The admin logs into the system using their credentials. 2. Upon successful login, the system displays the admin dashboard. 3. The admin selects an activity from the available list. 4. The system loads the relevant interface for the selected activity. 5. The admin proceeds with the task and confirms the action if necessary.
Alternative Flows:	If the admin lacks the required permissions for a specific task, the system displays an error message.
Exceptions:	None
Includes:	Verify User

Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	#7 – 2		
Use Case Name:	ManageUserAccounts		
Created By:	Eishani	Last Updated By:	Eishani
Date Created:	8 April 2025	Date Last Updated:	8 April 2025

Actor:	Admin
Description:	Admins manage user accounts, including viewing user roles (buyer or seller) and modifying account details.
Preconditions:	Admin must have valid login credentials and proper authorization.
Postconditions:	User accounts are updated successfully.
Priority:	High
Frequency of Use:	High
Flow of Events:	- Admin logs into the Admin Panel. - Admin navigates to the "Manage User Accounts" section. - Admin views user account details and updates them as necessary. - Admin confirms changes, and the system saves updates.
Alternative Flows:	If the admin lacks the required permissions for a specific task, the system displays an error message.
Exceptions:	None
Includes:	Verify Admin
Special Requirements:	None

Assumptions:	None
Notes and Issues:	None

4.7.3. Functional Requirements

7. Admin Management System

7.1. SG Homie Admin will be allowed to manage user accounts.

7.1.1. SG Homie provides an admin panel to manage user accounts.

7.1.2. Admins shall be able to view the status of the user, whether he/she is a seller or buyer.

7.2. SG Homie Admin will be able to manage properties.

7.2.1. Admins shall be able to manage all listed properties, including:

7.2.1.1.Editing property details

7.2.1.2.Removing inappropriate or fraudulent listings

7.2.2. Admins shall have the authority to approve or reject seller-submitted properties before they are published.

7.2.3. Admins shall be able to approve or reject new seller registrations based on the information provided.

7.2.4. Admins shall be able to review and verify property details submitted by sellers before publishing.

7.3. Admins shall be able to manage enquiries

7.3.1. Admins shall have access to all user-submitted enquiries through the admin panel.

7.3.2. Admins shall be able to view, manage, and reply to user enquiries directly from the system.

4.8. Security & Compliance

4.8.1. Functional Requirements

8. Security & Compliance

- 8.1. SG Hom-E shall implement secure authentication mechanisms.
 - 8.1.1. The system should use encryption for user passwords and personal data.
 - 8.1.2. The system shall follow PDPA guidelines for data privacy.
- 8.2. SG Hom-E shall maintain data integrity and prevent unauthorized access.
 - 8.2.1. The system shall restrict access to sensitive market data.
 - 8.2.2. The system shall log and monitor user activities to prevent abuse.

5. Other Nonfunctional Requirements

5.1. Performance Requirements

1. The SG Homie web application must load the homepage within 3 seconds to ensure a smooth user experience.
2. Property listings, transaction trends, and affordability insights should be displayed to users within 2 seconds for 95% of data retrieval operations.
3. Upon launching, the application should load and have all functionalities available within 5 seconds.
4. The platform must handle at least 1,000 concurrent users without significant performance degradation.

5.2. Safety Requirements

1. Only authorized developer team members shall access sensitive data related to property listings and user information.
2. The system shall perform automated daily backups to maintain data integrity and ensure quick recovery in case of data loss.

5.3. Security Requirements

1. The application must protect user privacy by securing all personal information obtained from users.
2. Users can only access their own saved searches, preferences, and personal data.
3. Ensure all property listings comply with Singapore's HDB and URA regulations.

5.4. Software Quality Attributes

1. The application should maintain 99.9% uptime, ensuring that services are available to users with minimal downtime.
2. The user interface should be intuitive, user-friendly, and readable, allowing users to navigate through the app and perform desired actions with ease.
3. The codebase should follow the open-closed principle, making it closed for modification but open for extension, facilitating easier addition of new functionalities.
4. The web application must be fully responsive, ensuring 100% of the content is visible and functional across various devices, including desktops and mobile devices, without content being clipped or hidden due to screen size limitations.

5.5. Business Rules

1. Users are assigned roles with specific permissions (e.g., regular user and admin). Each role has a different view of the web application and distinct permissions to ensure data security and proper management of functionalities.
2. Display disclaimers for AI-based predictions to avoid liability issues, informing users that predictions are estimates and should not be solely relied upon for critical decisions.

By adhering to these non-functional requirements, SG Homie aims to provide a reliable, secure, and user-friendly platform for users seeking housing information in Singapore.

6. Other Requirements

1. Legal Requirements: The application must comply with Singapore's PDPA.

7. Testing

7.1. Black-Box Testing

7.1.1. Account Registration

Test Scenario ID	Register-1		Test Case ID	Register-1A		
Test Case Description	Registration – Positive Test Case		Test Priority	High		
Pre-Requisite	Stable Internet Connection		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application		Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Try to find properties	The button “Start Exploring” should be clicked	Please sign in to access this feature	Please sign in to access this feature	Chrome	Pass
3	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass
4	User Sign Up for new users	The button “Sign up” should be clicked	Redirects to User Sign up page that requests for email and password	Redirects to User Sign up page that requests for email and password	Chrome	Pass
5	Sign up with name, email, and password that consists of less than 6 characters	Email, and password that is less than 6 characters.	An alert that says “Password cannot be less than 6 characters” at the bottom of the text input boxes.	An alert that says “Password cannot be less than 6 characters” at the bottom of the text input boxes.	Chrome	Pass
6	Sign up with name, email, and	Email, and password that	The landing page, as the user	The landing page, as the user	Chrome	Pass

	password that consists of 6 or more characters	consists of 6 or more characters.	registration is done successfully.	registration is done successfully.		
7	Sign up with existing name, email, and password that consists of 6 or more characters	Existing email, and password that consists of 6 or more characters.	Prompts “User already registered”	Prompts “User already registered”	Chrome	Pass
8	Completing Profile	Full Name, Monthly Income Range, Preferred Property Type, Preferred Location, and Number of Family Member entered	Prompts “Profile setup completed!”	Prompts “Profile setup completed!”	Chrome	Pass
9	Don’t complete profile after registration	Full Name, Monthly Income Range, Preferred Property Type, Preferred Location, or Number of Family Member entered	Couldn’t redirect to any page	Couldn’t redirect to any page	Chrome	Pass

7.1.2. Login Function

Test Scenario ID	Login-1		Test Case ID	Login-1A		
Test Case Description	Login – Positive Test Case		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass
3	Enter the email and password that have been registered	Registered Email and Password	Landing Page	Landing Page	Chrome	Pass

Test Scenario ID	Login-2		Test Case ID	Login-2A		
Test Case Description	Login – Negative Test Case		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass

3	Enter the email and password that have not been registered	Not registered Email	Shows “Invalid login credentials”	Shows “Invalid login credentials”	Chrome	Pass
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Test Scenario ID	Login-2	Test Case ID	Login-2B
Test Case Description	Login – Negative Test Case	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass
3	Enter the registered email, but wrong password	Email and Wrong Password	Shows “Invalid login credentials”	Shows “Invalid login credentials”	Chrome	Pass

Test Scenario ID	Login-3	Test Case ID	Login-3A
Test Case Description	Admin Login – Positive Test Case	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
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1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass
3	Click Sign in as Admin	The button “Sign in as Admin” should be clicked	Redirects to Admin Sign in page	Redirects to Admin Sign in page	Chrome	Pass
3	Enter the Admin Email and Password	Admin Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass

Test Scenario ID	Login-3	Test Case ID	Login-3B
Test Case Description	Admin Login – Negative Test Case	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass
3	Click Sign in as Admin	The button “Sign in as Admin” should be clicked	Redirects to Admin Sign in page	Redirects to Admin Sign in page	Chrome	Pass
3	Enter the Admin email, but wrong password	Email and Wrong Admin Password	Shows “Invalid login credentials”	Shows “Invalid login credentials”	Chrome	Pass

Test Scenario ID	Login-4		Test Case ID	Login-4A		
Test Case Description	Use Login – Negative Test Case		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Open Sign In page	The button “Sign in” should be clicked	Redirects to User Sign in page	Redirects to User Sign in page	Chrome	Pass
3	Click Sign in as Admin	The button “Sign in as Admin” should be clicked	Redirects to Admin Sign in page	Redirects to Admin Sign in page	Chrome	Pass
3	Enter the User email, but wrong password	User’s email and password	Shows “Access denied. Admin privileges required.”	Shows “Access denied. Admin privileges required.”	Chrome	Pass

7.1.3. Navigation Bar Functionalities

Test Scenario ID	Navbar-1		Test Case ID	Navbar-1A		
Test Case Description	Home Button		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Home” logo on the Navigation Bar	The Home button should be clicked	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass

Test Scenario ID	Navbar-2		Test Case ID	Navbar-2A		
Test Case Description	Start Exploring Button		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass

3	Click on the “Start Exploring” button on the Navigation Bar	The Start Exploring button should be clicked	Redirected to Search Properties page	Redirected to Search Properties page	Chrome	Pass
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Test Scenario ID	Navbar-3		Test Case ID	Navbar-3A		
Test Case Description	Explore Button		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Explore” button on the Navigation Bar	The Explore button is clicked	Redirected to Analytics Page	Redirected to Analytics Page	Chrome	Pass

Test Scenario ID	Navbar-4	Test Case ID	Navbar-4A
Test Case Description	Enquiry Button	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA
Test Execution Steps:			

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Enquiry” button on the Navigation Bar	The Explore button is clicked	Redirected to Enquiry page	Redirected to Enquiry Page	Chrome	Pass

Test Scenario ID	Navbar-4		Test Case ID	Navbar-4A		
Test Case Description	Become a Seller Button		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Become a Seller” logo on the Navigation Bar	The Become a Seller button is clicked. When the user is not a Seller yet, Become a seller button is displayed	Redirected to Seller Registration Page	Redirected to Seller Registration Page	Chrome	Pass

Test Scenario ID	Navbar-5		Test Case ID	Navbar-5A		
Test Case Description	Seller Dashboard Button (Upon becoming Seller)		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Seller Dashboard” logo on the Navigation Bar	The Seller Dashboard is clicked. When the user is already a seller, a seller dashboard button is displayed	Redirected to Seller Dashboard	Redirected to Seller Dashboard	Chrome	Pass

Test Scenario ID	Navbar-6		Test Case ID	Navbar-6A		
Test Case Description	Admin Button		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass

3	Click on the “Admin” logo on the Navigation Bar	The Admin is clicked. When the user is an admin, a admin dashboard button is displayed	Redirected to Admin Dashboard	Redirected to Admin Dashboard	Chrome	Pass
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7.1.4. Search Property Functionalities

Test Scenario ID	Search-1		Test Case ID	Search-1A		
Test Case Description	Get Recommendation from Profile		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Start Exploring” logo on the Navigation Bar	The Start Exploring should be clicked	Redirected to Property Page	Redirected to Property Page	Chrome	Pass
4	Click “Use recommendation” from profile	The Use Recommendation from profile should be clicked	Get the location and Room type from the user Profile	Get the location and Room type from the user Profile	Chrome	Pass
5	Unclick “Use recommendation” from profile	The Use Recommendation from profile should be unclicked	User can change the Location and Room Type	User can change the Location and Room Type	Chrome	Pass

Test Scenario ID	Search-2		Test Case ID	Search-2A		
Test Case Description	Property Filtering		Test Priority	High		

Pre-Requisite		Stable Internet Connection, an account that have been registered in the System		Post-Requisite		NA	
Test Execution Steps:							
No	Action	Inputs	Expected Output	Actual Output	Browser	Result	
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass	
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass	
3	Click on the “Start Exploring” logo on the Navigation Bar	The Start Exploring should be clicked	Redirected to Property Page	Redirected to Property Page	Chrome	Pass	
4	Filter the properties (inputing the Location, Number of Rooms, and price range)	Location, Number of Rooms, and Price Range are filleld	Display the properties based on those filters	Display the properties based on those filters	Chrome	Pass	
5	Unclick “Use recommend ation” from profile	The Use Recommendation from profile should be unclicked	User can change the Location and Room Type	User can change the Location and Room Type	Chrome	Pass	

7.1.5. Display Property Detail Functionalities

Test Scenario ID	Property-1		Test Case ID	Property-1A		
Test Case Description	Display Property Detail		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Start Exploring” logo on the Navigation Bar	The Start Exploring should be clicked	Redirected to Property Page	Redirected to Property Page	Chrome	Pass
4	Click on the one of the properties	One of the properties must be selected	Redirected to property details page	Redirected to property details page	Chrome	Pass

Test Scenario ID	Property-2		Test Case ID	Property-2A		
Test Case Description	Show Interest, Amenities Detail		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass

2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Start Exploring” logo on the Navigation Bar	The Start Exploring should be clicked	Redirected to Property Page	Redirected to Property Page	Chrome	Pass
4	Click on the one of the properties	One of the properties must be selected	Redirected to property details page, Nearby Amenities, Open Map should be displayed, Seller’s Information should be displayed	Redirected to property details page, Nearby Amenities, Open Map should be displayed, Seller’s Information should be displayed	Chrome	Pass
5	Click on “Show Interest”	Show Interest should be clicked	The show interest button should change into “Interested” and number of people interested should increase by 1	The show interest button should change into “Interested” and number of people interested should increase by 1	Chrome	Pass

7.1.6. Market Analytics Page Functionalities

Test Scenario ID	Analytics-1			Test Case ID	Analytics-1A	
Test Case Description	Analytics Page Functionalities			Test Priority	High	
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System			Post-Requisite	NA	
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Explore” button on the Navigation Bar	The Explore button should be clicked	Redirected to Market Analytics Page, and showing the Price Property Trend graph, including prediction, and Price comparison by town chart	Redirected to Market Analytics Page, and showing the Price Property Trend graph, including prediction, and Price comparison by town chart	Chrome	Pass

Test Scenario ID	Analytics-2		Test Case ID	Analytics-2A		
Test Case Description	Line Chart		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		

Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Explore” button on the Navigation Bar	The Explore button should be clicked	Redirected to Market Analytics Page, and showing the Price Property Trend graph, including prediction, and Price comparison by town chart	Redirected to Market Analytics Page, and showing the Price Property Trend graph, including prediction, and Price comparison by town chart	Chrome	Pass
4	Change the “ALL” filters on the Line Chart	The “All” should be changed into a year, for example 2019	The line chart for 2019’s trend only is displayed	The line chart for 2019’s trend only is displayed	Chrome	Pass

Test Scenario ID		Analytics-3		Test Case ID		Analytics-3A	
Test Case Description		Bar Chart		Test Priority		High	
Pre-Requisite		Stable Internet Connection, an account that have been registered in the System		Post-Requisite		NA	
Test Execution Steps:							
No	Action	Inputs	Expected Output	Actual Output	Browser	Result	

1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Explore” button on the Navigation Bar	The Explore button should be clicked	Redirected to Market Analytics Page, and showing the Price Property Trend graph, including prediction, and Price comparison by town chart	Redirected to Market Analytics Page, and showing the Price Property Trend graph, including prediction, and Price comparison by town chart	Chrome	Pass
4	Click on “Select Town” and select other Towns	The “Select Towns” should be change accordingly into 5 different towns, e.g. Ang Mo Kio, Bedok, Bishan, Bukit Batok, Bukit Merah	The Bar Chart for the 5 selected properties should be displayed	The Bar Chart for the 5 selected properties should be displayed	Chrome	Pass

7.1.7. Enquiries Page Functionalities

Test Scenario ID	Enquiries-1		Test Case ID	Enquiries-1A		
Test Case Description	Enquiry Page Functionalities		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Enquiry” button on the Navigation Bar	The Enquiry button should be clicked	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Chrome	Pass

Test Scenario ID	Enquiries-2	Test Case ID	Enquiries-2A
Test Case Description	Send Enquiry – Positive Test Results	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Enquiry” button on the Navigation Bar	The Enquiry button should be clicked	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Chrome	Pass
4	Fill all of the blank fields using desired format	The Full Name, Phone Number, and Message should be filled, and the Send Message Button should be clicked	The Enquiry will be displayed at the right side, and PENDING status is shown	The Enquiry will be displayed at the right side, and PENDING status is shown	Chrome	

Test Scenario ID	Enquiries-2	Test Case ID	Enquiries-2B
Test Case Description	Send Enquiry – Negative Test Results	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Enquiry” button on the Navigation Bar	The Enquiry button should be clicked	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Chrome	Pass
4	Fill the blank fields using desired format, but don’t fill the Name	Phone Number, and Message should be filled, the Name should be left empty, and the Send Message Button should be clicked	The website will gives out “Please fill out this field” message.	The website will gives out “Please fill out this field” message.	Chrome	Pass

Test Scenario ID	Enquiries-2	Test Case ID	Enquiries-2C
Test Case Description	Send Enquiry – Negative Test Results	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA
Test Execution Steps:			

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Enquiry” button on the Navigation Bar	The Enquiry button should be clicked	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Redirected to Enquiry Page, and showing the enquiry fields, that asks to input Full Name, Phone Number, and their message to be sent	Chrome	Pass
4	Fill in all of the empty fields, but write the Phone Number in the wrong formatting	Name, Phone Number, and Message should be filled, the Phone Number is written in the wrong format, and the Send Message Button should be clicked	The website will gives out “Please match the requested format” message.	The website will gives out “Please match the requested format” message.	Chrome	Pass

7.1.8. Seller Registration Page Functionalities

Test Scenario ID	SelllerRegistration-1		Test Case ID	SellerRegistration-1A		
Test Case Description	Seller Registration Page Functionalities		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Become a Seller” button on the Navigation Bar	The Become a Seller button should be clicked	Redirected to Seller Registration Page, and shows a Seller Registration Form	Redirected to Seller Registration Page, and shows a Seller Registration Form	Chrome	Pass

Test Scenario ID	SelllerRegistration-2		Test Case ID	SellerRegistration-2A		
Test Case Description	Seller Form – Positive Test Cases		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass

2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Become a Seller” button on the Navigation Bar	The Become a Seller button should be clicked	Redirected to Seller Registration Page, and shows a Seller Registration Form	Redirected to Seller Registration Page, and shows a Seller Registration Form	Chrome	Pass
4	Fill in the Phone Number, and Check the I agree to the terms and condition	The Phone number should be filled, and I agree to terms and conditions should be ticked, and Register as a Seller Button checked	Redirected to Seller Dashboard page	Redirected to Seller Dashboard Page	Chrome	Pass

Test Scenario ID	SelllerRegistration-2	Test Case ID	SellerRegistration-2B
Test Case Description	Seller Form – Negative Test Cases	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Become a Seller” button on the	The Become a Seller button should be clicked	Redirected to Seller Registration Page, and	Redirected to Seller Registration Page, and	Chrome	Pass

	Navigation Bar		shows a Seller Registration Form	shows a Seller Registration Form		
4	Fill in the Phone Number, and I agree to the terms and condition is not checked	The Phone number should be filled, and I agree to terms and conditions should not be ticked	Register as Seller is not click-able	Register as Seller is not click-able	Chrome	Pass

7.1.9. Seller Dashboard Page Functionalities

Test Scenario ID	SellerDashboard-1		Test Case ID	SellerDashboard-1A		
Test Case Description	Seller Dashboard Page Functionalities		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Seller Dashboard” button on the Navigation Bar	The Seller Dashboard button should be clicked, and had registered as a seller previously	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass

Test Scenario ID		SellerDashboard-2		Test Case ID		SellerDashboard-2A	
Test Case Description		Add Property Functionalities		Test Priority		High	
Pre-Requisite		Stable Internet Connection, an account that have been registered in the System		Post-Requisite		NA	
Test Execution Steps:							
No	Action	Inputs	Expected Output	Actual Output	Browser	Result	

1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Seller Dashboard” button on the Navigation Bar	The Seller Dashboard button should be clicked, and had registered as a seller previously	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on “Add Properties” button on the seller dashboard	The Add Property button should be clicked	Showing a form for Adding a new property	Showing a form for Adding a new property	Chrome	Pass
5	Fill in the information for the Property	The information, such as Title, Price, Location, Detailed Location, Bedrooms, Bathrooms, Area, Built Year, Seller Name, Contact Number, Property Main Images, and Descriptions should be filled, and Add Property button should be clicked	Showing “Property added successfully”, the Property is shown in the Seller Dashboard with status of PENDING	Showing “Property added successfully”, the Property is shown in the Seller Dashboard with status of PENDING	Chrome	Pass

Test Scenario ID	SellerDashboard-2	Test Case ID	SellerDashboard-2B
Test Case Description	Add Property Functionalities – Negative Test Case	Test Priority	High

Pre-Requisite		Stable Internet Connection, an account that have been registered in the System		Post-Requisite		NA	
Test Execution Steps:							
No	Action	Inputs	Expected Output	Actual Output	Browser	Result	
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass	
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass	
3	Click on the “Seller Dashboard” button on the Navigation Bar	The Seller Dashboard button should be clicked, and had registered as a seller previously	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass	
4	Click on “Add Properties” button on the seller dashboard	The Add Property button should be clicked	Showing a form for Adding a new property	Showing a form for Adding a new property	Chrome	Pass	
5	Fill in the informations for the Property, but leave 1 not following the format	The informations, such as Title, Price, Location, Detailed Location, Bedrooms, Bathrooms, Area, Built Year, Seller Name, Contact Number, Property Main Images, and Descriptions should be filled, the	The website will gives out “Please match the requested format” message.	The website will gives out “Please match the requested format” message.	Chrome	Pass	

		number of Bedrooms is filled with “A”, and Add Property button should be clicked				
--	--	--	--	--	--	--

Test Scenario ID	SellerDashboard-2	Test Case ID	SellerDashboard-2C
Test Case Description	Add Property Functionalities – Negative Test Case	Test Priority	High
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA

Test Execution Steps:

No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Seller Dashboard” button on the Navigation Bar	The Seller Dashboard button should be clicked, and had registered as a seller previously	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on “Add Properties” button on the seller dashboard	The Add Property button should be clicked	Showing a form for Adding a new property	Showing a form for Adding a new property	Chrome	Pass
5	Fill in the informations for	The informations, such as Price,	The website will gives	The website will gives out	Chrome	Pass

	the Property, but leave 1 field blank	Location, Detailed Location, Bedrooms, Bathrooms, Area, Built Year, Seller Name, Contact Number, Property Main Images, and Descriptions should be filled, the Title of the property should be left blank, and Add Property button should be clicked	out “Please fill out this field” message.	“Please fill out this field” message.		
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Test Scenario ID	SellerDashboard-3			Test Case ID	SellerDashboard-3A	
Test Case Description	Edit Property Functionalities			Test Priority	High	
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System			Post-Requisite	NA	
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Seller Dashboard” button on the Navigation Bar	The Seller Dashboard button should be clicked, and had registered as a seller previously	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the	Chrome	Pass

			property registration	property registration		
4	Click on the “Edit” logo, beside the property on the seller dashboard	The Edit button should be clicked	Showing a form for editing the property	Showing a form for editing the property	Chrome	Pass
5	Fill in the required fields	Edit the required fields, and click on Update Properties	Redirect back to Seller Dashboard pages, and it should reflect the new property’s detail	Redirect back to Seller Dashboard pages, and it should reflect the new property’s detail	Chrome	Pass

7.1.10. Admin Dashboard Page Functionalities

Test Scenario ID	AdminDashboard-1		Test Case ID	AdminDashboard-1A		
Test Case Description	Admin Dashboard Page Functionalities		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass

Test Scenario ID		AdminDashboard-2		Test Case ID		AdminDashboard-2A	
Test Case Description		Manage Users		Test Priority		High	
Pre-Requisite		Stable Internet Connection, an account that have been registered in the System		Post-Requisite		NA	
Test Execution Steps:							
No	Action	Inputs	Expected Output	Actual Output	Browser	Result	

1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on the “Manage User”	The Manage Users button should be clicked	Showing the list of users, with their Name, Email, Seller Status, and number of their properties sold	Showing the list of users, with their Name, Email, Seller Status, and number of their properties sold	Chrome	Pass

Test Scenario ID		AdminDashboard-3	Test Case ID		AdminDashboard-3A	
Test Case Description		Approve Properties	Test Priority		High	
Pre-Requisite		Stable Internet Connection, an account that have been registered in the System	Post-Requisite		NA	
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result

1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on the “Manage Properties”	The Manage Properties button should be clicked	Showing the list of Pending Properties and Approved Properties	Showing the list of Pending Properties and Approved Properties	Chrome	Pass
5	Click on the Check Box on the pending properties	The tick should be checked	The property now should move into Accepted property and shown in the Search Property page	The property now should move into Accepted property and shown in the Search Property page	Chrome	Pass

Test Scenario ID	AdminDashboard-3	Test Case ID	AdminDashboard-3B
Test Case Description	Reject Properties	Test Priority	High

Pre-Requisite		Stable Internet Connection, an account that have been registered in the System		Post-Requisite		NA	
Test Execution Steps:							
No	Action	Inputs	Expected Output	Actual Output	Browser	Result	
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass	
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass	
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass	
4	Click on the “Manage Properties”	The Manage Properties button should be clicked	Showing the list of Pending Properties and Approved Properties	Showing the list of Pending Properties and Approved Properties	Chrome	Pass	
5	Click on the Check Box on the pending properties	The cross should be checked	The property now should move into Rejected property	The property now should move into Rejected property	Chrome	Pass	

Test Scenario ID	AdminDashboard-3	Test Case ID	AdminDashboard-3C			
Test Case Description	Delete Properties	Test Priority	High			
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA			
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on the “Manage Properties”	The Manage Properties button should be clicked	Showing the list of Pending Properties and Approved Properties	Showing the list of Pending Properties and Approved Properties	Chrome	Pass
5	Click on the Trash Bin icon on the Approved properties	The trash bin icon should be checked	The property should now be deleted, and not shown in the Search Property Page	The property should now be deleted, and not shown in the Search Property Page	Chrome	Pass

Test Scenario ID	AdminDashboard-4	Test Case ID	AdminDashboard-4A			
Test Case Description	Reply Enquiry	Test Priority	High			
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA			
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on the “Manage Enquiry”	The Manage Enquiry button should be clicked	Showing the list of enquiry with their status	Showing the list of enquiry with their status	Chrome	Pass
5	Click on the Reply button on the Pending Enquiries	The reply button should be clicked	The reply box should prompt out	The reply box prompted out	Chrome	Pass
6	Type in the reply in the Reply form	The required field must be filled	Redirect to Manage Enquiry page and shows REPLIED	Redirect to Manage Enquiry page and shows REPLIED	Chrome	Pass

Test Scenario ID	AdminDashboard-4	Test Case ID	AdminDashboard-4B			
Test Case Description	Delete Enquiry	Test Priority	High			
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA			
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Admin” button on the Navigation Bar	The Admin button should be clicked, and had registered as a admin previously	Redirected to a Admin Dashboard, with list of total users, pending properties, Approved properties, and Total Enquiries	Redirected to a Seller Dashboard, with list of properties that he/she sells before, and the status of the property registration	Chrome	Pass
4	Click on the “Manage Enquiry”	The Manage Enquiry button should be clicked	Showing the list of enquiry with their status	Showing the list of enquiry with their status	Chrome	Pass
5	Click on the trash bin icon on the Enquiries	The trash bin button should be clicked	The enquiry should be deleted	The enquiry should be deleted	Chrome	Pass

7.1.11. Profile Management Functionalities

Test Scenario ID	Profile-1	Test Case ID	Profile-1A			
Test Case Description	Profile Dashboard Functionalities	Test Priority	High			
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA			
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Profile” button on the Navigation Bar	The profile button must be clicked	Redirect to a profile page, which shows the Name of User, Email of User, and details of the User Profile, with Edit Profile button and Sign Out button	Redirect to a profile page, which shows the Name of User, Email of User, and details of the User Profile, with Edit Profile button and Sign Out button	Chrome	Pass

Test Scenario ID	Profile-2	Test Case ID	Profile-2A			
Test Case Description	Edit Profile	Test Priority	High			
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System	Post-Requisite	NA			
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass

2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Profile” button on the Navigation Bar	The profile button must be clicked	Redirect to a profile page, which shows the Name of User, Email of User, and details of the User Profile, with Edit Profile button and Sign Out button	Redirect to a profile page, which shows the Name of User, Email of User, and details of the User Profile, with Edit Profile button and Sign Out button	Chrome	Pass
4	Click on Edit Profile	The edit profile button should be clicked	The user profile fields are now editable	The user profile fields are now editable	Chrome	Pass
5	Click on Save Changes	The save changes button should be clicked	Redirects to profile page, with new profile data updated	Redirects to profile page, with new profile data updated	Chrome	Pass

Test Scenario ID	Profile-3		Test Case ID	Profile-3A		
Test Case Description	Sign Out		Test Priority	High		
Pre-Requisite	Stable Internet Connection, an account that have been registered in the System		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Sign in	Registered Email and Password	Redirected to Landing Page	Redirected to Landing Page	Chrome	Pass
3	Click on the “Profile”	The profile button must be clicked	Redirect to a profile page,	Redirect to a profile page,	Chrome	Pass

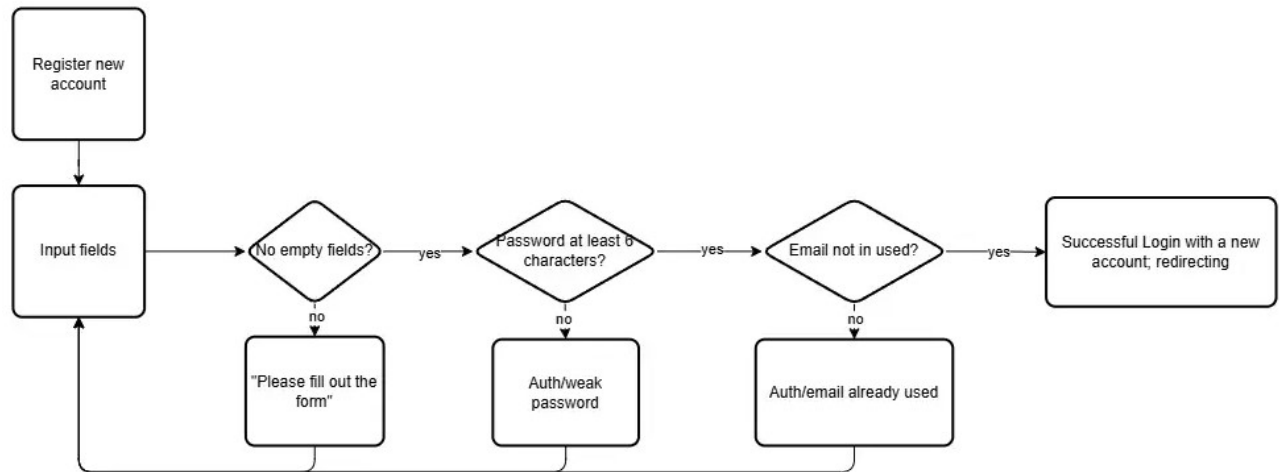
	button on the Navigation Bar		which shows the Name of User, Email of User, and details of the User Profile, with Edit Profile button and Sign Out button	which shows the Name of User, Email of User, and details of the User Profile, with Edit Profile button and Sign Out button		
4	Click on Sign Out	The Sign out button should be clicked	The user are now logged out, and redirects to the Landing Page	The user are now logged out, and redirects to the Landing Page	Chrome	Pass

7.1.12. Chatbot Functionalities

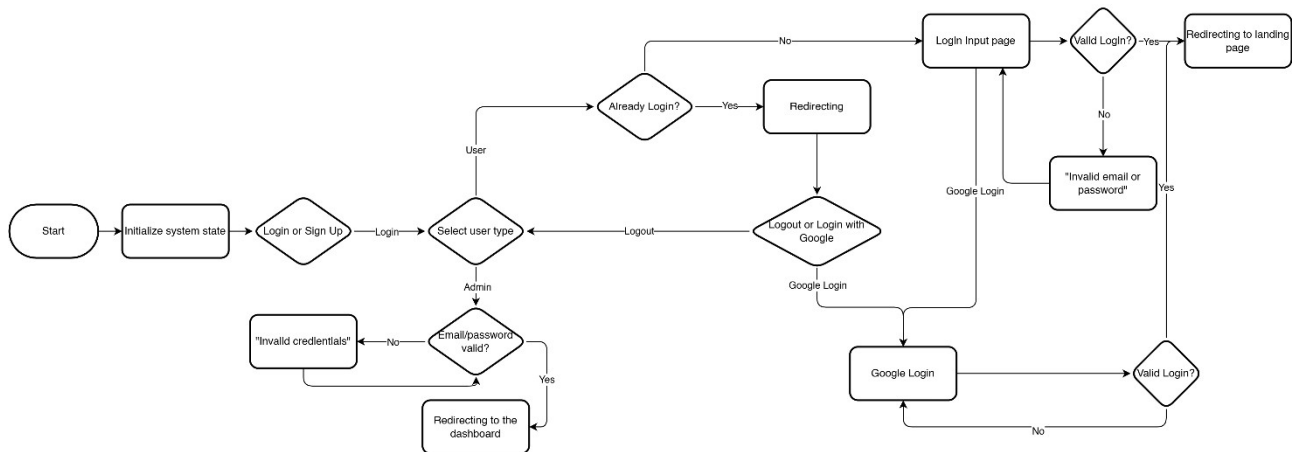
Test Scenario ID	Bot-1		Test Case ID	Bot-1A		
Test Case Description	Chatbot Functionalities		Test Priority	High		
Pre-Requisite	Stable Internet Connection		Post-Requisite	NA		
Test Execution Steps:						
No	Action	Inputs	Expected Output	Actual Output	Browser	Result
1	Launch Application	https://sg-homie.netlify.app/	Initial Landing Page	Initial Landing Page	Chrome	Pass
2	Click on the Chat Bot icon at the bottom right of the page	The ChatBot should be clicked	Opens a small window of chatbot	Opens a small window of chatbot	Chrome	Pass
3	Click on one of the option in the chatbot	One of the option should be clicked, e.g. Support	Replies with how to contact support	Replies with “You can reach our support team through: Email: support@sghomie.com Phone: +65 6789 0123 Contact form on our website Our support hours are Mon-Fri, 9am-6pm.”	Chrome	Pass
4	Click on “X” at the top right of the chatbot	The Close button should be clicked	Closes the chatbot	Closes the chatbot	Chrome	Pass

7.2. White Box Testing

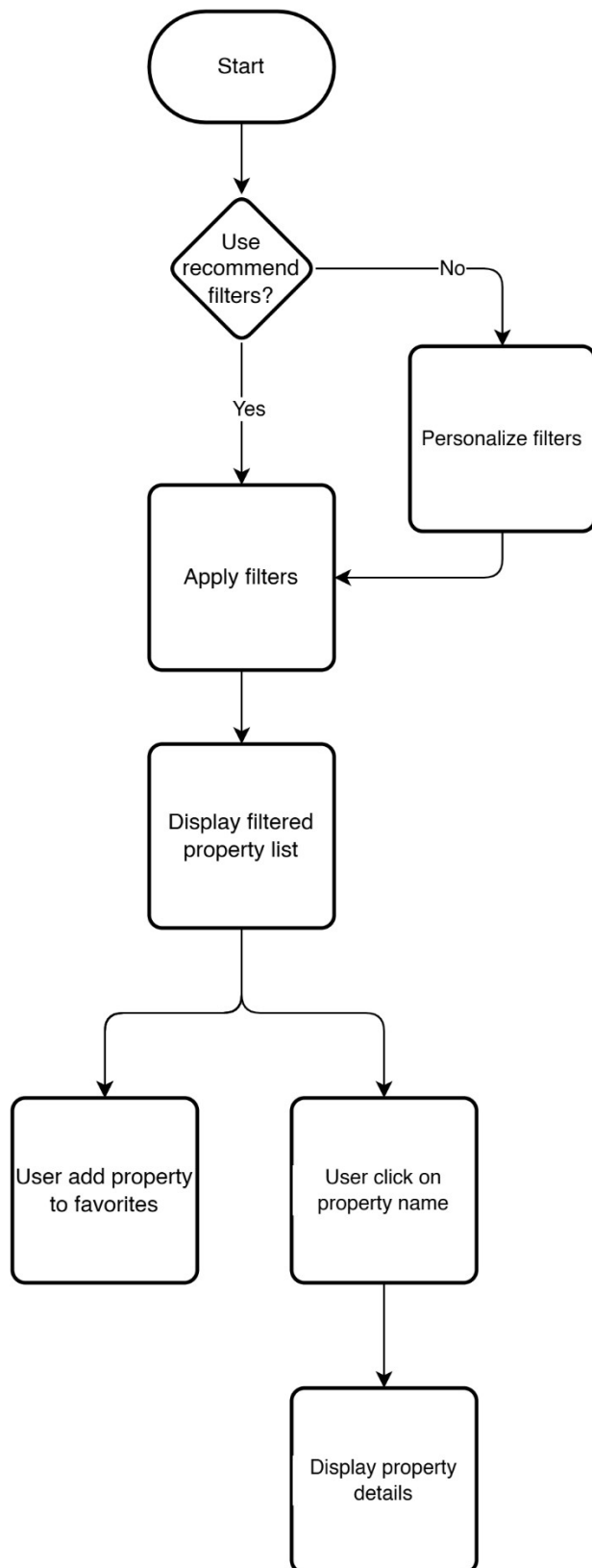
7.2.1. User Sign Up



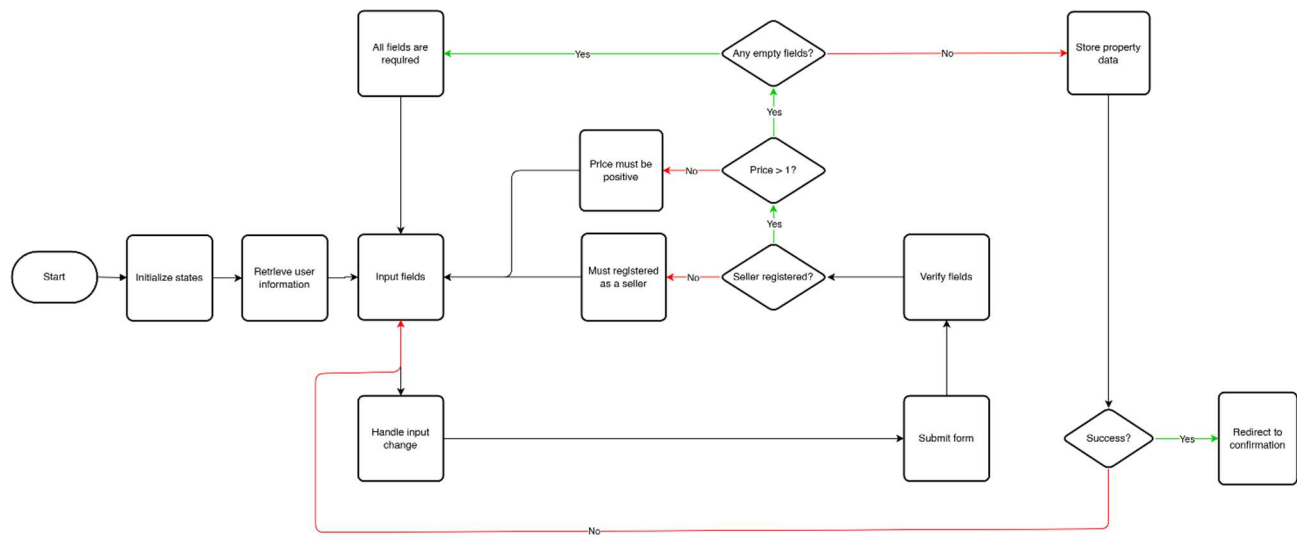
7.2.2. User Log In



7.2.3. Property Search



7.2.4. Add Property



Appendix A: Glossary / Data Dictionary

Account

A user profile registered on SG-Homie that stores personal information, housing preferences, saved property searches, and notification settings.

Admin

A user with administrative privileges who oversees platform management, content moderation, and verification of property-related data.

Application (App)

The SG-Homie web-based application designed for both desktop and mobile browsers, enabling access to real estate analytics, housing tools, and personalized recommendations.

Buyer

An individual utilizing SG-Homie to explore and evaluate residential properties for potential purchase using tools such as price trend analysis and affordability calculators.

Seller

A property owner leveraging SG-Homie's features to analyze market performance and determine competitive pricing for their listings.

Investor

A user focused on identifying high-potential property investments, relying on SG-Homie's data-driven insights, forecasts, and regional market trends.

Property Listing

A comprehensive display of property attributes including location, pricing, floor area, tenure, and transaction history curated for platform users.

Housing Estate

A geographical residential cluster in Singapore, either public (HDB) or private, managed by appropriate authorities and commonly analyzed on the platform.

Resale Transaction

A completed property sale recorded in public databases (HDB, URA), used by SG-Homie to derive insights into historical pricing and market trends.

Affordability Calculator

An interactive tool that estimates a user's purchasing capacity based on income, CPF contributions, loan eligibility, and down payment capabilities.

Market Sentiment

An aggregated interpretation of real estate market dynamics, reflecting trends in transaction volume, price fluctuations, and user engagement.

Project Pipeline

A catalog of upcoming residential developments including Build-To-Order (BTO) projects, private condominiums, and government land sales.

Price Forecasting

Artificial intelligence-powered predictive analytics that estimate future housing prices using historical data, economic indicators, and urban development trends.

Neighborhood Insights

A feature providing contextual information on amenities, transportation links, education institutions, and lifestyle factors within specific housing locales.

Map

An interactive geospatial interface that visualizes property listings, transaction data, and points of interest to aid user navigation and decision-making.

OneMap API

A third-party geolocation service that enables SG-Homie to calculate travel distances, accessibility, and surrounding infrastructure for property listings.

User Profile Collection

The process of capturing user preferences and requirements, enabling the platform to generate personalized property recommendations.

Scoring System

An AI-driven evaluation model that ranks properties based on metrics such as affordability, accessibility, and investment potential.

Notification System

An automated alert mechanism that informs users of updates on property listings, price changes, market trends, and regulatory news.

Saved Search

A feature allowing users to bookmark customized property filters for quick access and automatic update tracking.

Government Regulation Compliance

The platform's adherence to regulatory standards set by Singapore's HDB and URA authorities for legal and ethical data presentation.

AI Prediction Disclaimer

A disclaimer indicating that predictions made by SG-Homie's AI tools are estimates based on historical data and should not be taken as financial advice.

Session Timeout

A security measure that automatically logs users out after a period of inactivity to protect personal data and account integrity.

Heatmap

A visual representation of areas with high real estate activity, enabling users to identify popular housing zones and emerging trends.

Data Refresh Interval

The frequency with which platform data is updated to reflect the latest property information and ensure accuracy.

Cross-Device Synchronization

The capability that allows users to access synchronized account data seamlessly across multiple devices.

Frontend

The client-facing component of SG-Homie that encompasses user interface design and user experience functionality.

Backend

The server-side architecture responsible for data processing, logic execution, and integration with databases and APIs.

Latency

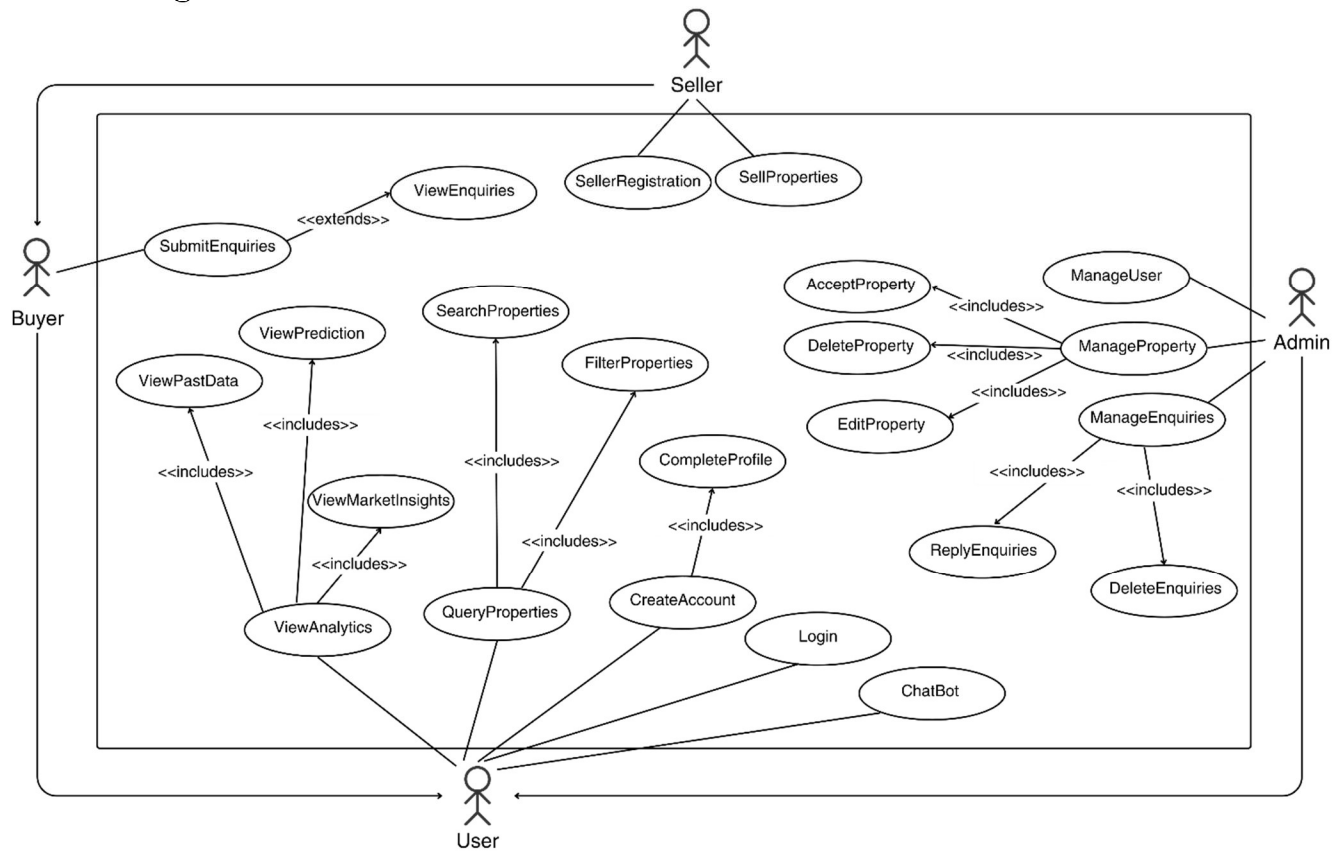
The response time between a user's action and the platform's reaction. SG-Homie aims to maintain minimal latency for a seamless experience.

Cookie Consent

A user agreement acknowledging the use of cookies to enhance site functionality and user personalization, in compliance with privacy laws.

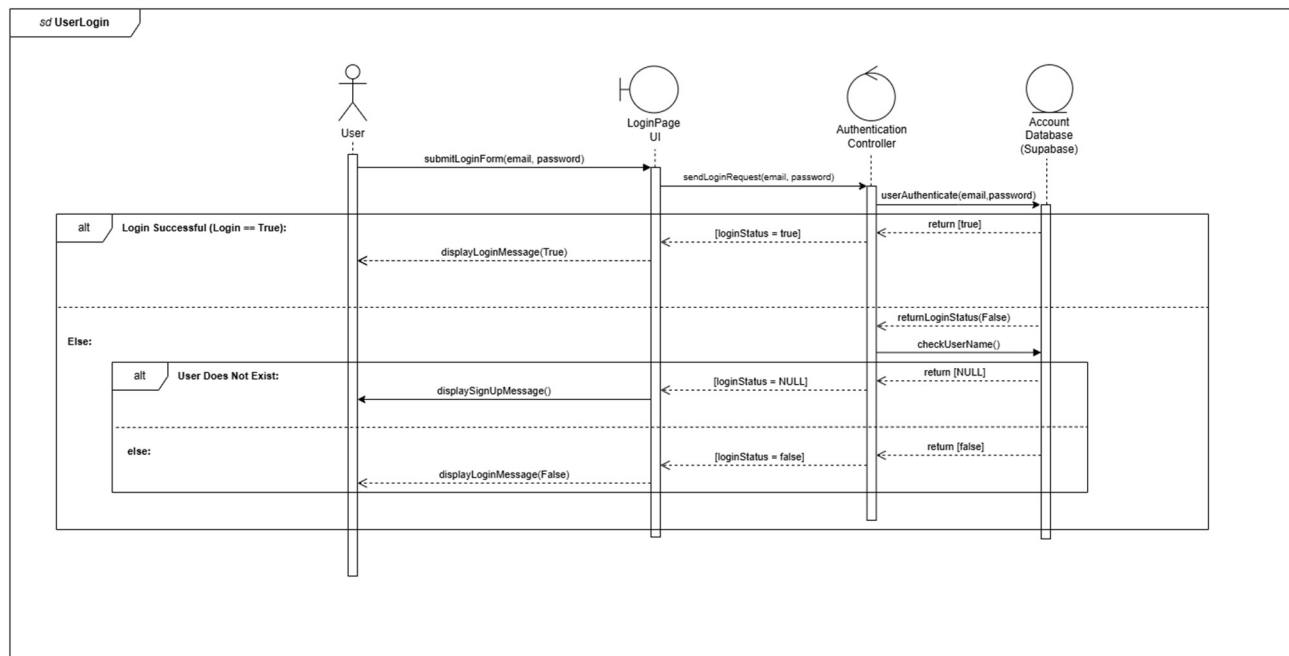
Appendix B: Analysis Models

Use Case Diagram

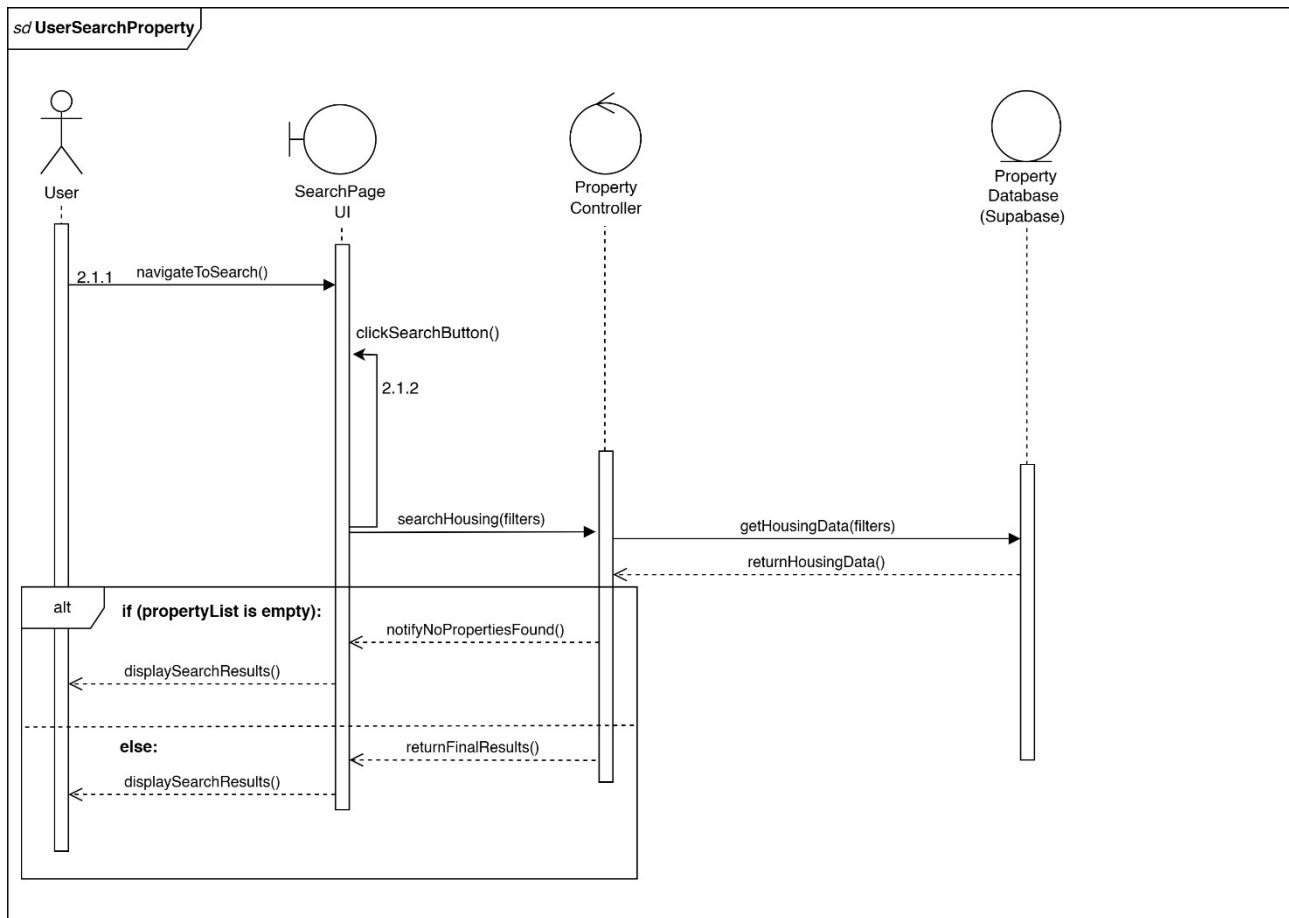


Sequence Diagram

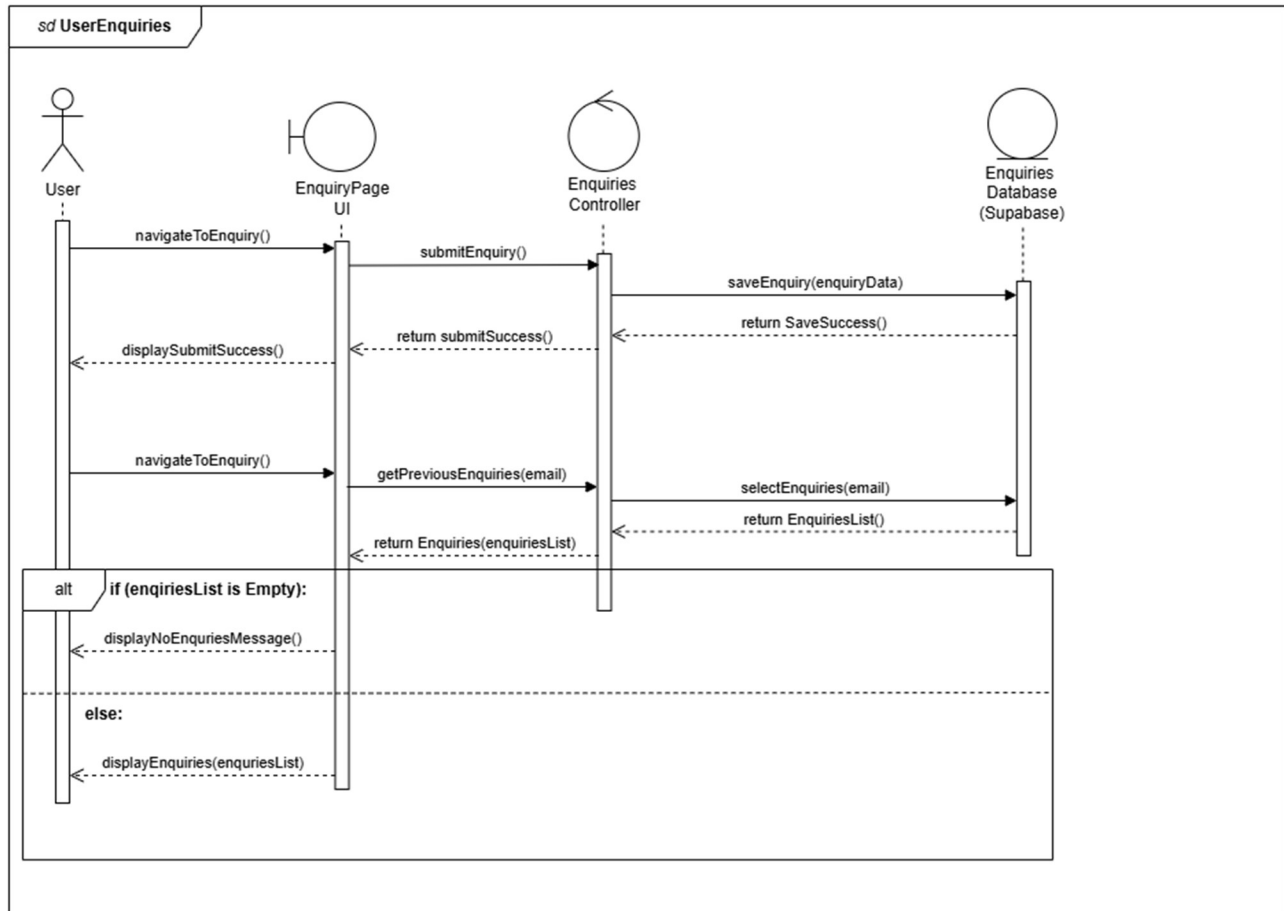
Use Case 1: User Login and Completing Profile



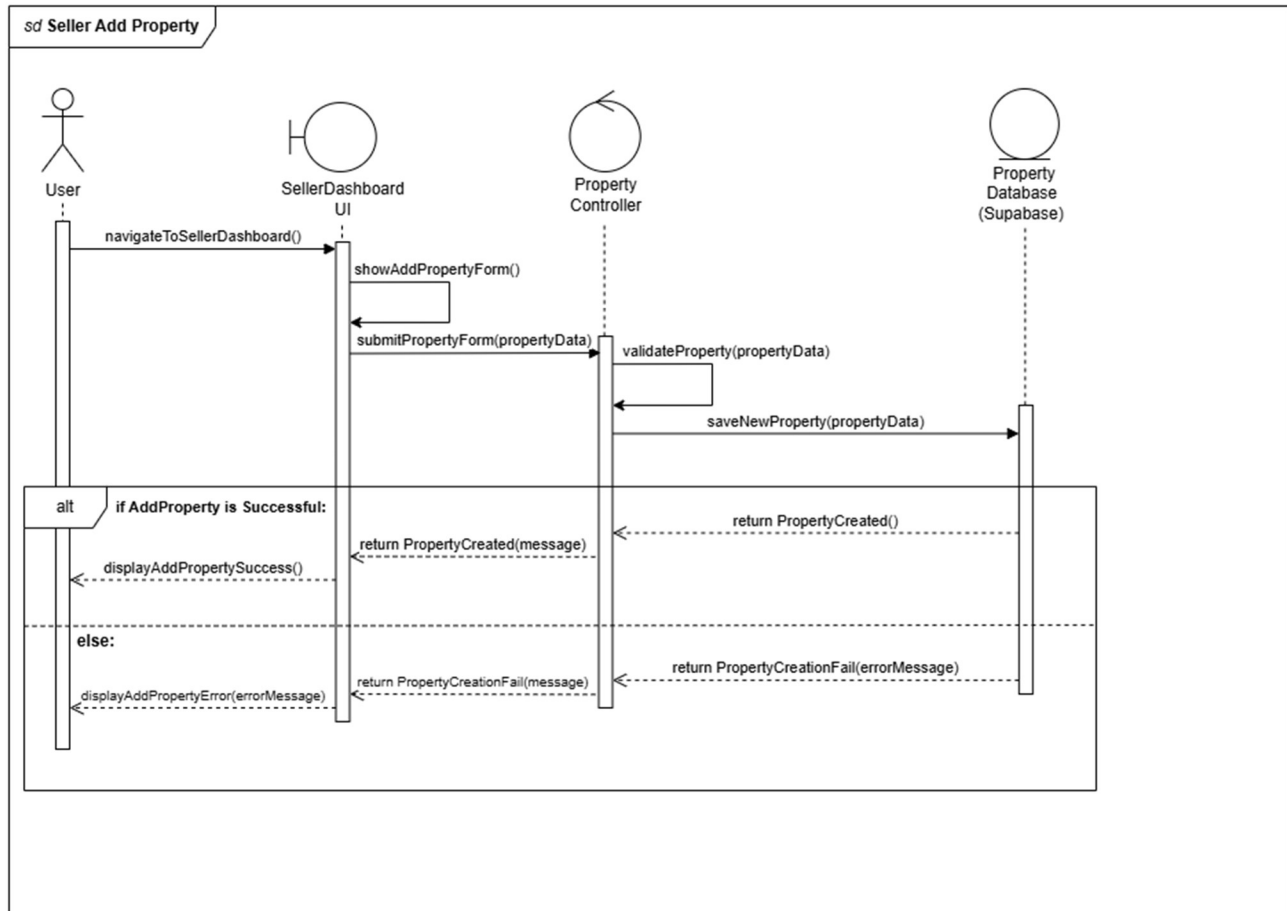
Use Case 2: Search Property



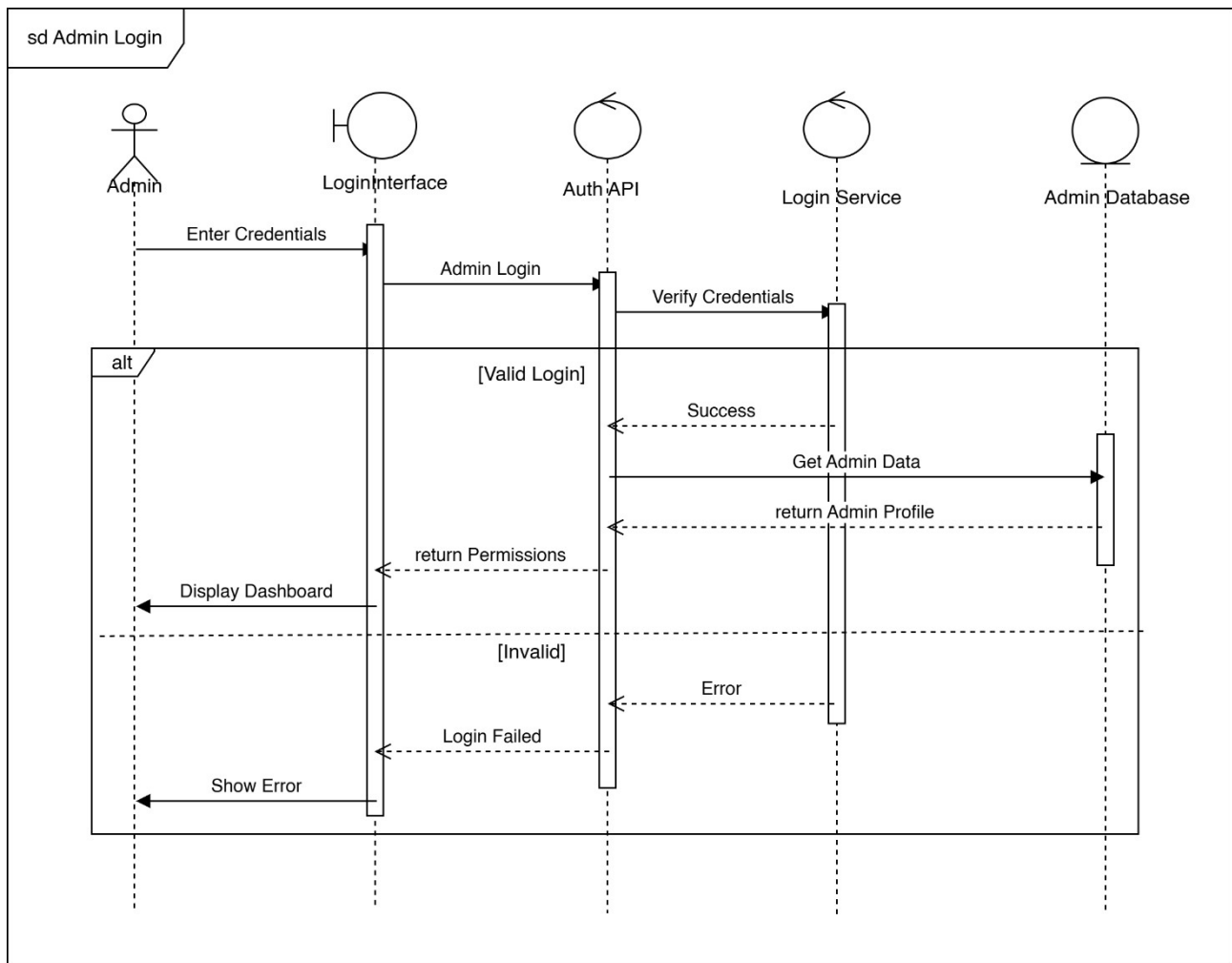
Use Case 3: User Enquiries



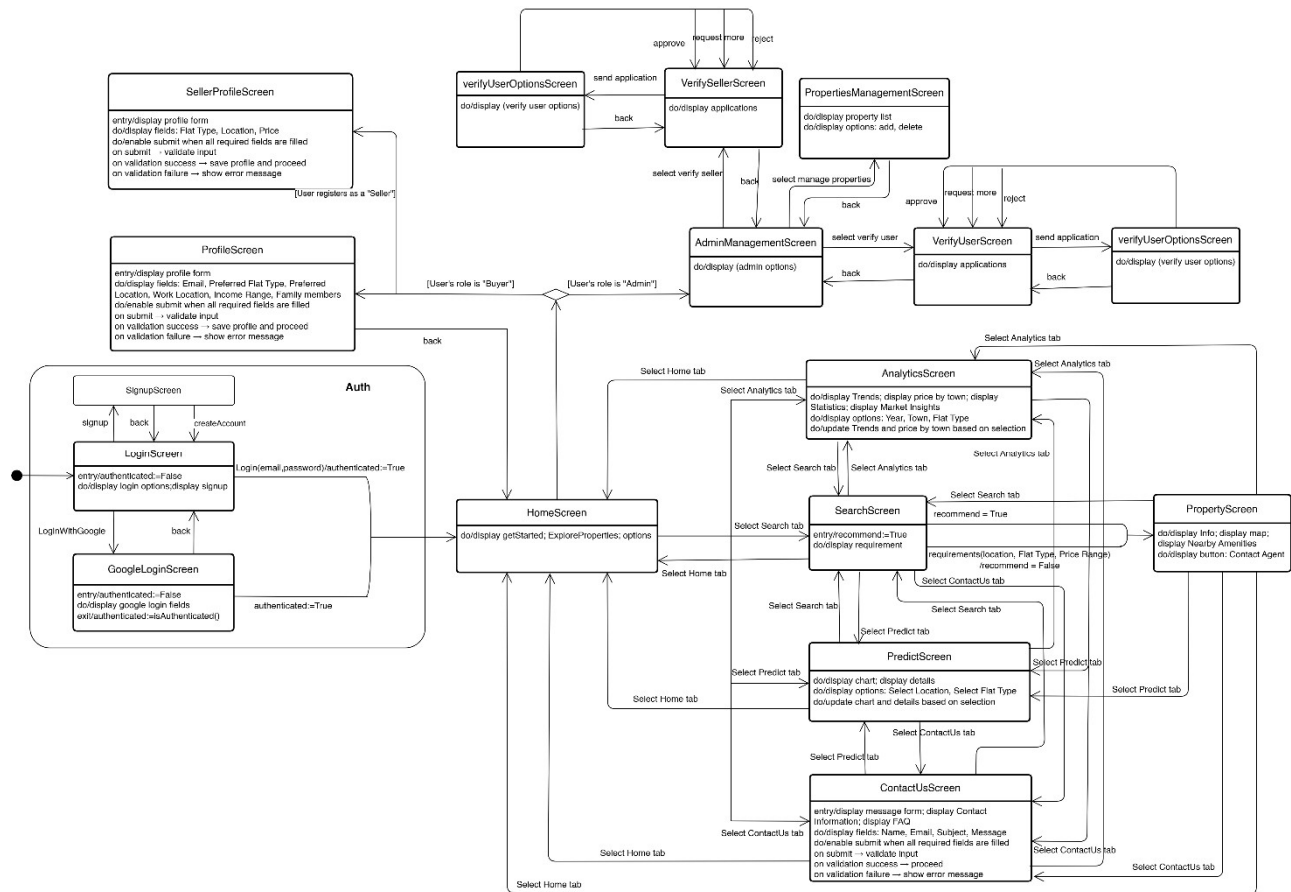
Use Case 4: Seller Add Properties



Use Case 5: Admin Login

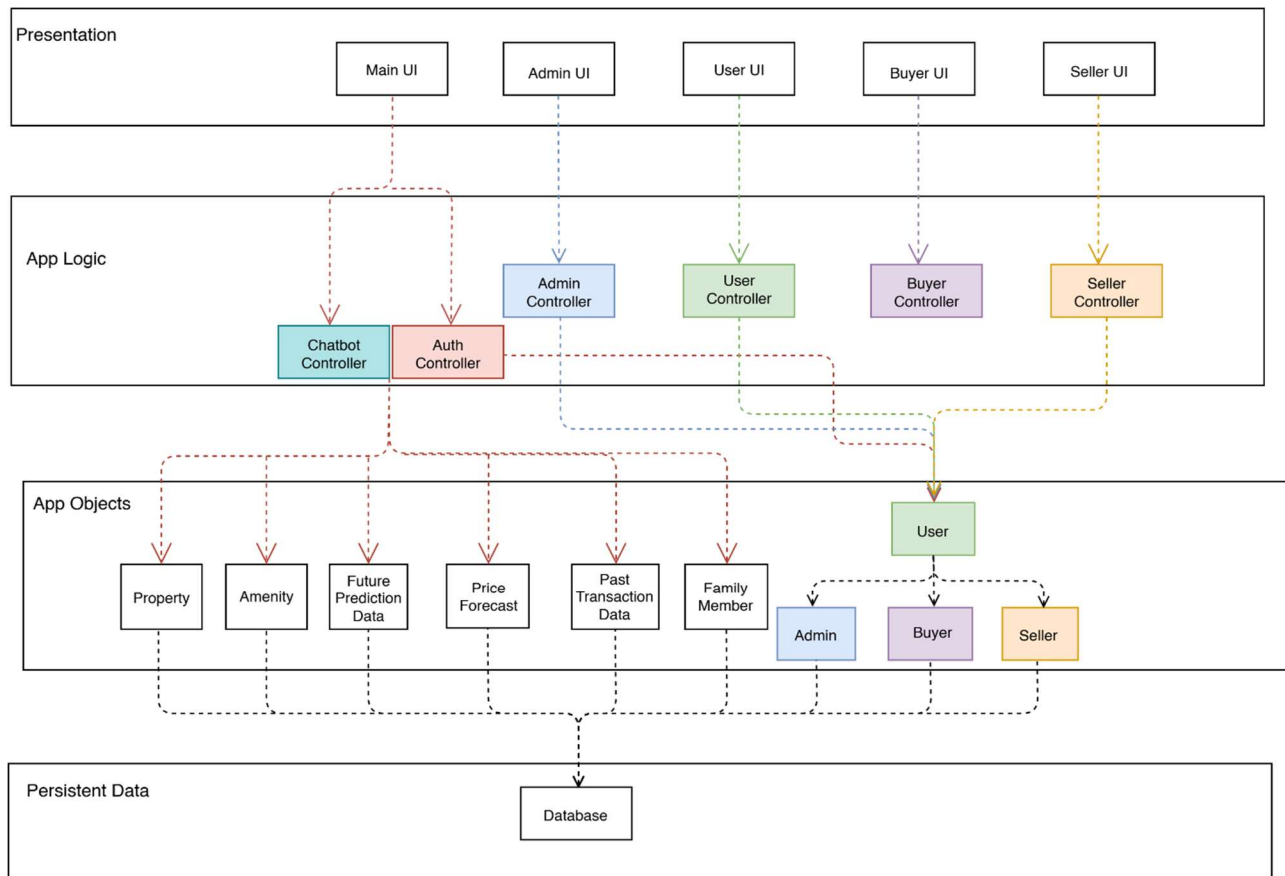


Dialog Map



Please refer to the attached files for clearer image.

System Architecture



Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>

Source: http://www.frontiernet.net/~kwiegers/process_assets/srs_template.doc